

Chapter 2: Open Source Intelligence

Definition, History, and Practice

Companion Workbook to the Instructional Video

How to Use This Workbook

This is the written companion to the instructional video on open source intelligence. It works three ways: as a follow-along guide while you watch, as a standalone textbook chapter if you never watch, and as a practice manual, since roughly half of it is exercises.

This chapter assumes you have covered the material in Chapter 1 (information versus intelligence, the intelligence lifecycle, the collection disciplines, and the strategic/operational/tactical levels). Where Chapter 1 concepts reappear, they are briefly restated so you are never stuck, but the review questions assume you know them.

The audience assumption is the same as Chapter 1: you can think in systems and follow a technical process, but you have not worked in intelligence, journalism, or investigations. Every specialized term is defined the first time it appears and again in Key Terms.

Time estimate:

Component	Estimated time
Video	45 to 60 minutes
Reading material	4 to 5 hours
Knowledge assessment	1 hour
Critical thinking exercises	1.5 to 2 hours
Labs 1 through 5	9 to 14 hours total
Capstone assignment	5 to 7 hours
Total	21 to 30 hours

Materials you will need: a computer with internet access, a browser, a notebook or word processor, and a documentation habit. No paid tools are required for any lab in this chapter. Every technique taught here can be performed with a browser and patience.

A Warning Before You Begin

This chapter teaches you to find information about people, organizations, and events using public

sources. That capability is genuinely useful and genuinely dangerous, and the difference between the two is entirely in how you use it.

Three rules govern every exercise in this workbook. They are not suggestions and they are graded.

Rule 1: No private individuals. Every lab and the capstone use fictional entities, historical public events, or organizations. You will not be asked to research a real private person, and doing so on your own initiative as part of this coursework is grounds for a zero regardless of the quality of your work. A classmate, an ex, a neighbor, a person from the news who is not a public figure: all off limits.

Rule 2: Publicly available means genuinely public. If you have to create a false identity, defeat an access control, guess a password, use someone else's credentials, or accept a document someone was not authorized to share, you have left OSINT and entered something else. The something else has criminal statutes attached to it in most jurisdictions.

Rule 3: Document as you go. Every finding gets a source, a URL or citation, a date, and a timestamp. This is not busywork. It is what separates an investigator from a person with opinions, and it is the only thing that will protect you if a finding is ever challenged.

If a lab ever seems to require breaking one of these rules, you have misread the lab. Stop and re-read it.

Learning Objectives

By the end of this chapter you should be able to do all of the following.

Knowledge objectives (you can explain it):

1. Define open source intelligence and state precisely what makes a source "open."
2. Distinguish open source information from open source intelligence.
3. Describe the major eras in the development of OSINT, from press monitoring through the current period.
4. Explain the technological and social changes that drove OSINT from a supporting activity to a primary discipline.
5. Explain why OSINT matters, in terms of cost, speed, shareability, and access.
6. Describe how OSINT is integrated into modern all-source intelligence work.
7. Analyze historical intelligence failures and identify the specific failure mode in each.
8. Analyze historical intelligence successes and identify what OSINT specifically contributed.
9. Describe how governments, commercial organizations, and private individuals each use OSINT, and how their constraints differ.
10. State the legal and ethical boundaries that separate OSINT from unlawful collection, harassment, and privacy violation.

Application objectives (you can do it):

1. Identify open source information available from everyday sources.
2. Convert a set of open source findings into an intelligence product with a judgment, a confidence level, and named gaps.
3. Evaluate a source for reliability and a specific claim for credibility, as separate judgments.
4. Detect circular reporting by tracing a claim toward its origin.
5. Corroborate a claim across genuinely independent sources.
6. Document a collection trail that another person could audit and reproduce.
7. Produce a professional OSINT report on an organization using only public information.

Analysis objectives (you can judge it):

1. Determine whether OSINT is the appropriate discipline for a given requirement, or whether the requirement cannot be answered from open sources at all.
 2. Recognize deception, manipulation, and coordinated inauthentic content in open sources.
 3. Assess an intelligence failure and specify what better collection or analysis would have changed, distinguishing failures of collection from failures of analysis and from failures of dissemination.
 4. Apply an ethical test to a proposed collection activity before performing it.
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Part 1: Reading Material

1.1 The Definition of OSINT

1.1.1 The Working Definition

Open source intelligence (OSINT) is intelligence produced from information that is publicly available, lawfully obtainable, and collected without breaching any protection the owner placed on it.

Unpack that in three parts, because each one does work.

"Intelligence." OSINT is not a pile of search results. Carrying forward the Chapter 1 definition, intelligence is information collected against a stated requirement, evaluated for reliability, analyzed in context, given a judgment and a confidence level, and delivered to a decision maker in time to matter. Everything in Chapter 1 applies here. OSINT names where the raw material came from; it does not lower the bar for what makes the output intelligence.

"Publicly available." The information can be obtained by any member of the public through lawful means. It does not have to be easy to find. It does not have to be free. It does not have to be indexed by a search engine. A court record buried in a county clerk's paper archive is publicly available. A \$4,000

commercial satellite image is publicly available. A corporate registry that charges a fee is publicly available.

"Without breaching any protection." This is the boundary. If the owner took a measure to restrict access and you worked around that measure, the information is no longer open, regardless of how weak the measure was. An unlocked door is still a door.

1.1.2 OSIF and OSINT: A Distinction That Matters

Practitioners distinguish several layers, and the vocabulary is worth learning because it prevents a common professional embarrassment.

Layer	Name	What it is	Example
1	Open Source Data (OSD)	Raw, unprocessed public material as it exists	A raw satellite image; an unedited livestream; a full court docket PDF
2	Open Source Information (OSIF)	Public material that has been collected, collated, and given basic context	A news article; a compiled list of a company's filings; a summarized docket
3	Open Source Intelligence (OSINT)	OSIF that has been evaluated, analyzed against a requirement, and given a judgment	"This company will likely open a Southeast facility in Q4. High confidence."
4	Validated OSINT (OSINT-V)	OSINT confirmed against non-open sources or with very high certainty	An OSINT judgment corroborated by classified reporting

The professional embarrassment: delivering Layer 2 while calling it Layer 3. A 40-page compilation of everything publicly known about a company is OSIF. It becomes OSINT when someone reads it and says what it means, what happens next, and how confident they are. Chapter 1's ladder applies without modification.

1.1.3 What Counts as an Open Source

The category is wider than most students expect. A partial inventory:

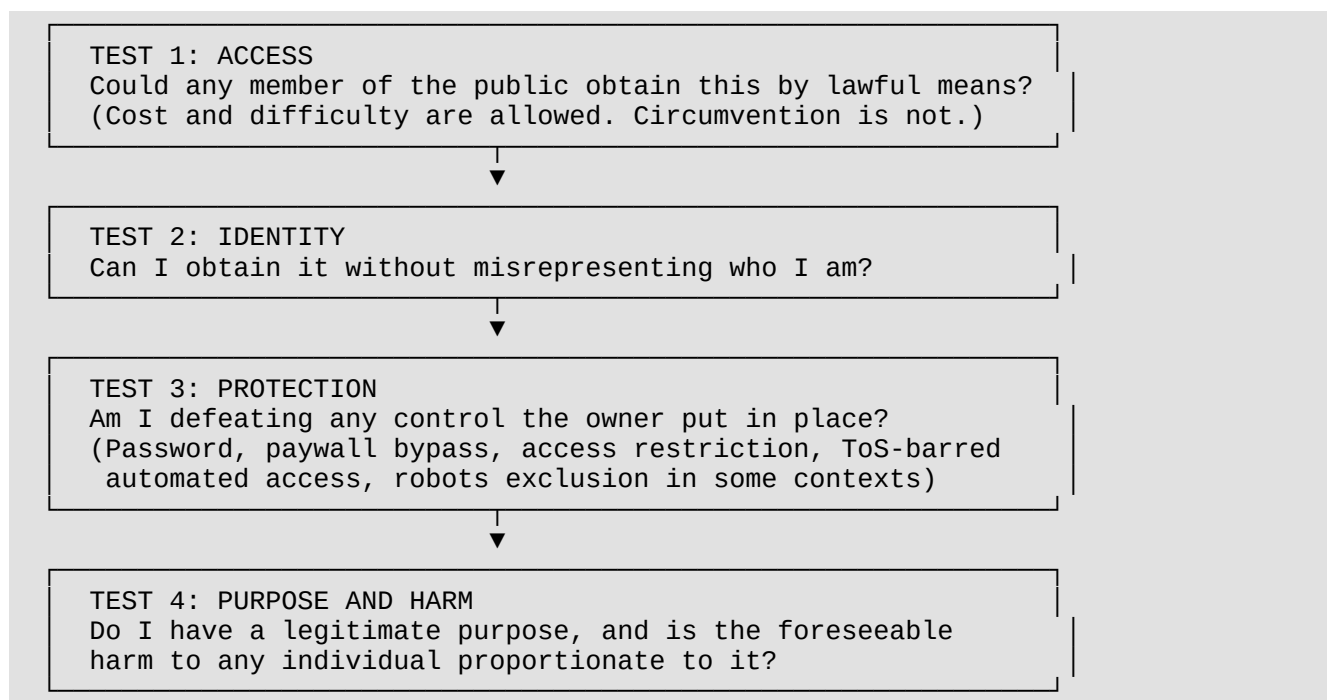
Category	Examples
News and media	Newspapers, broadcast, wire services, trade press, local outlets, foreign-language media
Academic and technical	Journals, conference proceedings, theses, preprint servers, technical standards
Government publications	Statistics, budgets, regulatory filings, inspection reports, legislative records, hearings, procurement

Category	Examples
	notices
Legal records	Court dockets, judgments, liens, bankruptcies, property records, licensing boards
Corporate records	Registrations, annual reports, investor calls, patents, trademarks, UCC filings
Geospatial	Commercial satellite imagery, aerial photography, street-level imagery, terrain and map data
Transportation	Ship tracking (AIS), aircraft tracking (ADS-B), rail and freight publications, flight schedules
Technical internet	Domain registration records, certificate transparency logs, DNS records, internet-wide scan data, public code repositories
Social and user-generated	Social platforms, forums, review sites, video platforms, hobbyist communities, livestreams
Commercial data	Market research, credit reports where lawfully accessible, industry databases, subscription data services
Gray literature	Conference slides, internal-looking documents posted publicly, marketing collateral, job postings
Human open sources	Public conference talks, published interviews, expert commentary, and conversations where you identify yourself honestly

That last row surprises people. A conversation at a trade show, where you say who you are and ask a question anyone could ask, is open source collection. It is only HUMINT in the specialized sense when it involves a cultivated source relationship or elicitation.

1.1.4 The Four Tests for "Open"

Before collecting anything, run it through four questions. If any answer is no, stop.



Test 4 is the one that separates a professional from a technician. The first three tests are about legality. The fourth is about judgment. Aggregating twenty individually public facts about a private person into a single profile can produce something that person never made public and would be harmed by, and the fact that each input was public does not resolve that. This is sometimes called the **mosaic effect**, and it is the central ethical problem of the discipline.

1.1.5 Common Misconceptions

"OSINT means free information." No. It means lawfully accessible information. Commercial satellite imagery, subscription databases, and court record retrieval fees are all normal OSINT expenses.

"OSINT means internet research." No. The internet is the largest open source channel, not the definition. Print archives, municipal records rooms, library holdings, and public meetings are open sources, and they are frequently the ones nobody else checked.

"If it's on the internet, it's fair game." No. A document accidentally exposed on a misconfigured server was not published. A private profile you accessed through a fake account was not public to you. Availability is not the same as authorization.

"OSINT is easier than other disciplines." No. It is cheaper and lower-risk. It is not easier. The volume, the deception, and the verification burden make competent OSINT analytically demanding in a way that a single authoritative sensor feed is not.

"OSINT is a lesser discipline." This was once a common institutional view and it has substantially collapsed. Section 1.3 explains why.

1.2 The History of OSINT

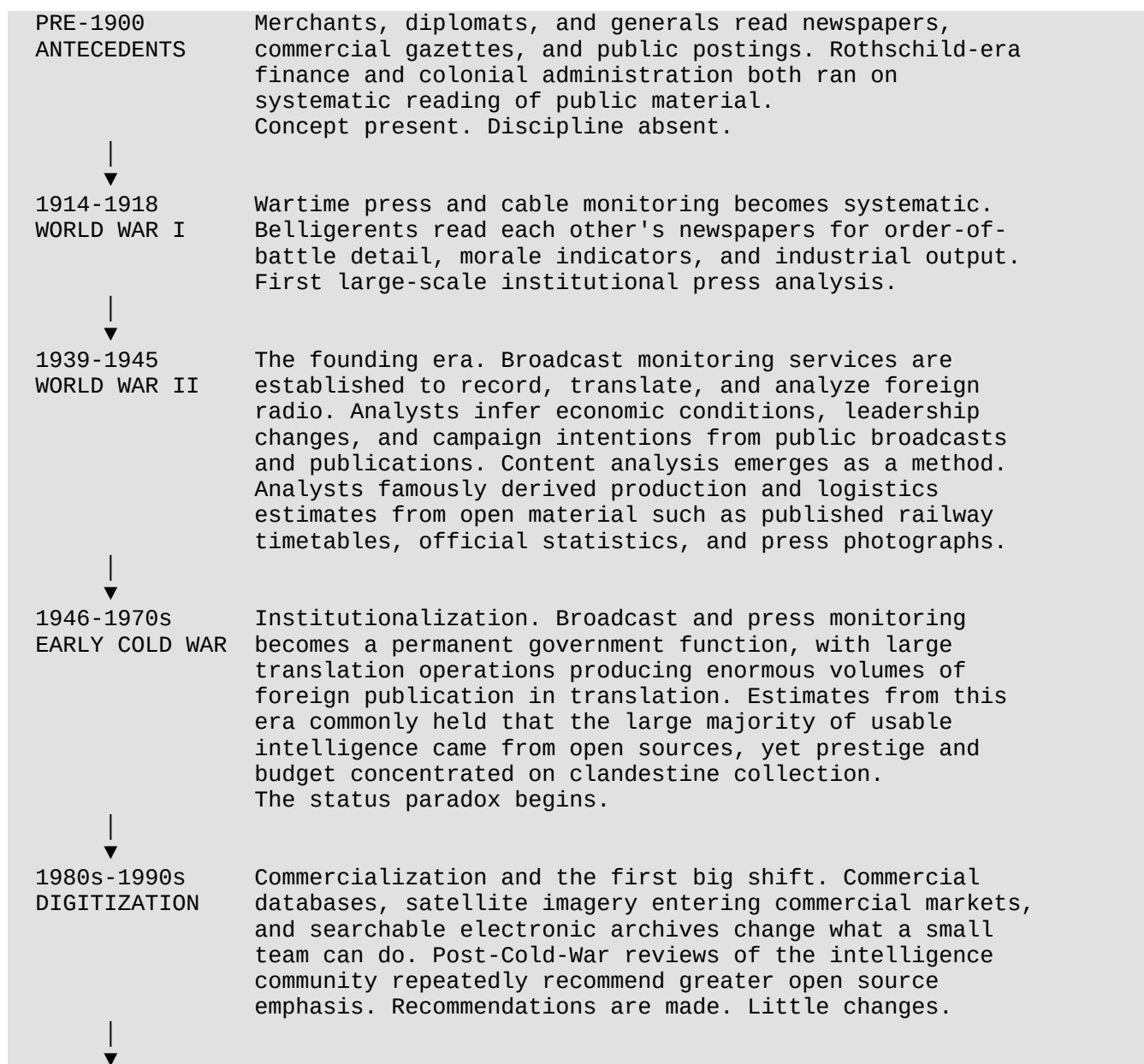
1.2.1 Why History Matters Here

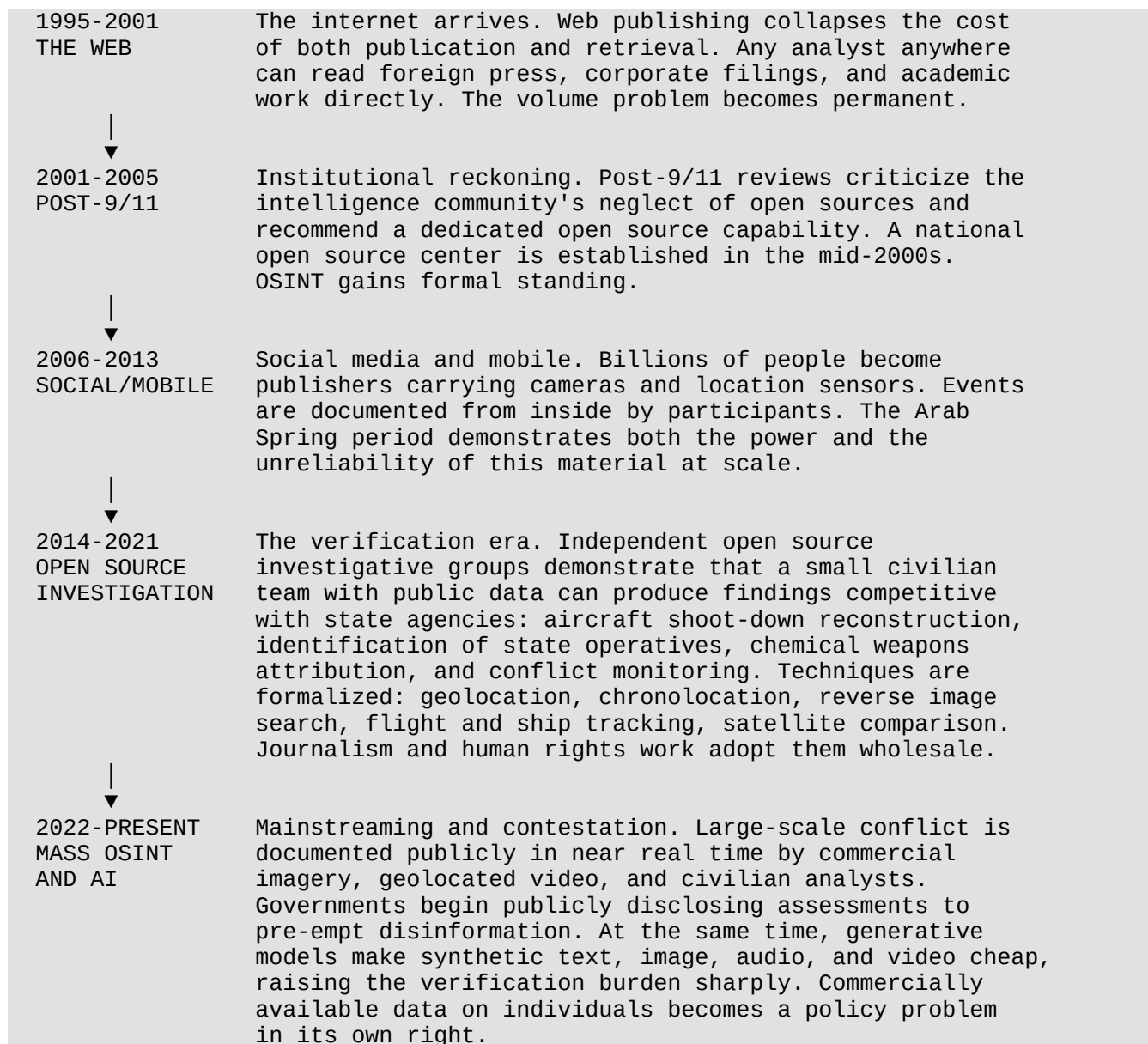
Two reasons to study this timeline rather than skim it.

First, every recurring problem in modern OSINT appeared earlier in a lower-technology form. Information overload was a real crisis for press-monitoring services in the 1940s. Deception through open channels is as old as newspapers. Circular reporting predates the internet by centuries. Knowing the earlier version helps you recognize the current one.

Second, OSINT's institutional status has swung back and forth repeatedly. Understanding that cycle explains why organizations still underinvest in it despite decades of evidence.

1.2.2 The Timeline





1.2.3 Four Recurring Lessons From the Timeline

Lesson 1: The value was recognized early and the investment lagged anyway. For most of the post-1945 period, credible internal assessments held that open sources supplied most of the usable content of finished intelligence, while budget and prestige went to secret collection. The gap persisted for decades. If you work in this field, expect to make this argument again.

Lesson 2: Every capability expansion brought a proportional deception risk. Broadcast monitoring invited planted broadcasts. Web publishing invited fabricated sites. Social media invited coordinated inauthentic behavior. Generative models invite synthetic evidence. The pattern is reliable enough to plan around: assume that any new open channel will be exploited for deception within a short period of becoming influential.

Lesson 3: The bottleneck moved from access to verification. In 1943 the hard part was obtaining a

foreign broadcast. In 2026 the hard part is establishing that a video showing a specific event was actually filmed at the claimed place and time. Collection got easy. Verification got hard. Your training should be weighted accordingly, and this chapter's labs are weighted that way deliberately.

Lesson 4: Capability decentralized. In 1960, meaningful open source analysis at scale required a government. By the mid-2010s, a small team with a browser could contribute findings of genuine national significance. That decentralization is why this chapter covers personal use alongside government and commercial use. It is also why ethics carries more weight than it used to: capability without institutional oversight is a new condition.

1.3 The Evolution of OSINT

1.3.1 What Actually Changed

The history section gives the sequence. This section gives the mechanism. Six changes converted OSINT from a clipping service into a primary discipline.

- 1. Publication cost collapsed.** When publishing required a printing press or a transmitter, the population of publishers was small and institutional. When it required a phone, the population became most of humanity. The consequence for analysts: events are now documented from the inside by participants, not only from the outside by correspondents.
- 2. Sensors became consumer goods.** A modern phone contains a camera, a satellite positioning receiver, an accelerometer, a compass, and a clock, and it embeds their output in files. Every participant in an event is a potential sensor platform, whether they intend to be or not.
- 3. Commercial imagery matured.** Satellite imagery moved from a state monopoly to a purchasable commodity with revisit rates measured in days or hours. A newsroom or a hedge fund can now buy what once required a national reconnaissance program.
- 4. Machine-readable public infrastructure appeared.** Aircraft and vessels broadcast their identity and position by regulation. Certificate transparency logs record every publicly trusted certificate issued. Corporate and legal registries went online. Internet-wide scanning services publish what is exposed. None of this was designed as an intelligence resource, and all of it functions as one.
- 5. Verification methods were formalized.** Geolocation from visual features, chronolocation from shadows and weather, reverse image search, and cross-referencing against satellite archives were developed into teachable, repeatable techniques. This is the single most important professional development in the discipline's recent history, because it made open source findings defensible to skeptical audiences: courts, editors, and adversarial governments.
- 6. The community became public.** Techniques are documented in the open, taught in the open, and critiqued in the open. Errors get caught by strangers. That adversarial peer review is a genuine methodological advantage over closed systems, and it is unusual among intelligence disciplines.

1.3.2 The Evolution in One Table

Dimension	Pre-1990	1990s to 2000s	2010s	Current period
Dominant sources	Print, broadcast, official publications	Web pages, commercial databases	Social media, mobile video, commercial imagery	All of the above plus machine-readable feeds and inferred data
Primary constraint	Access to material	Retrieval and search	Volume and triage	Verification and authenticity
Who can do it	Governments and large institutions	Institutions and well-funded firms	Small teams and skilled individuals	Nearly anyone, with widely varying quality
Time to first finding	Weeks	Days	Hours	Minutes, with proportionally higher error risk
Dominant skill	Language and subject expertise	Search and database skill	Verification and geolocation	Authentication, provenance, and calibrated judgment
Dominant failure mode	Missing the material	Missing the signal in the volume	Believing unverified material	Believing synthetic material, and over-collecting on individuals
Institutional status	Low prestige, high contribution	Recommended, underfunded	Formally recognized	Central, and increasingly regulated

1.3.3 The Verification Burden, Illustrated

To make the shift concrete, compare the work required to establish one claim across eras.

Claim: a military convoy moved through a specific town on a specific date.

Era	How you would establish it	Effort	Residual uncertainty
1943	Monitor local broadcast, read regional press, correlate with railway timetables	Weeks; requires linguists	High; press was censored
1985	Request national reconnaissance tasking; wait for imagery and interpretation	Days to weeks; requires a state	Low, but access limited to a few
2005	Search web reporting,	Days	Moderate; reporting

Era	How you would establish it	Effort	Residual uncertainty
	corroborate across outlets, check commercial imagery if affordable		may be circular
Now	Locate participant video, geolocate from visible features, chronolocate from shadow angle and weather records, corroborate against same-day satellite imagery, verify upload metadata and account history, check for prior appearances of the same footage	Hours, by one person	Low if done properly; high if any step is skipped

Read the last row carefully. The modern method is faster, cheaper, more distributed, and more rigorous *when performed correctly*, and it is worse than useless when performed carelessly, because it produces confident conclusions from recycled footage. The technique list in that cell is the actual content of modern OSINT tradecraft, and Labs 3 and 5 will have you practice parts of it.

1.3.4 What Has Not Changed

Three things survive every technological shift, which is why they get the most emphasis in this workbook.

The requirement still comes first. Better tools did not make aimless collection productive. They made it faster to accumulate irrelevant material.

Source evaluation is still two questions. Reliability of the source and credibility of the specific claim, judged separately. Chapter 1's framework applies unchanged to a satellite image and to a forum post.

A judgment still has to be made by a person who is accountable for it. No amount of collection produces a conclusion on its own. Someone has to say what it means and own being wrong.

1.4 Why OSINT Matters

1.4.1 The Seven Structural Advantages

These are properties of the discipline, not opinions about it.

1. Cost. Most open source collection costs analyst time and little else. Compare that to running a

human source network or building a sensor. This changes who can participate: a two-person team, a newsroom, a county sheriff's office, a mid-sized company.

2. Speed. Open sources are often the first available reporting on a developing event, because participants publish faster than institutions collect. In the opening hours of an incident, open sources may be the only sources.

3. Shareability. This is the most underrated advantage. OSINT findings carry no classification and no source-protection burden, so they can be shared with partners, allies, tenants, customers, the press, a jury, or the public. A classified finding that cannot be acted on publicly is often worth less than an unclassified finding that can. Governments have increasingly disclosed open-source-based assessments publicly precisely because disclosure is possible.

4. Accessibility. No clearance is required to read it, produce it, or consume it. That means the analyst pool is not limited to cleared personnel, and the consumer pool is not limited to cleared officials.

5. Auditability. An open source finding can be reproduced by a skeptic. The URL, the image, the filing, the timestamp: all of it can be independently checked. No other discipline offers this. It is why open source evidence has become usable in legal and quasi-legal proceedings.

6. Tipping and cueing. OSINT frequently identifies where to point expensive collection. A public shipping record or a job posting can tell you which facility to image or which question to ask, converting a broad search into a narrow one. In a resource-constrained program, this is often OSINT's highest-value function.

7. Breadth. Open sources cover topics no classified system was built for: local politics, market sentiment, technical standards, public health behavior, supply chain conditions, weather, infrastructure. For most questions an ordinary organization actually has, open sources are not the supporting discipline. They are the only discipline.

1.4.2 The Six Structural Limitations

Honesty about these is a professional obligation, and it is graded in this chapter.

1. Volume. The material vastly exceeds any team's capacity to read it. Triage is not optional and triage discards things.

2. Deception. Anyone can publish. Publishing is the cheapest way to reach an analyst, which makes open channels the preferred vector for deliberate deception. Assume that any topic important enough to be collected on is important enough for someone to seed.

3. Circular reporting. Apparent corroboration that traces back to one unverified origin. Endemic in fast-moving events and the single most common cause of confident open source error.

4. Availability bias. Open sources over-represent what is easy to publish from: connected populations, permissive environments, English-language material, urban areas, and events that photograph well. Absence of open reporting is weak evidence of absence.

5. It rarely reaches intent. Open sources show behavior, capability, and stated position. What a

decision maker privately plans, and what the internal disagreement is, generally remains outside open reach. Chapter 1's point stands: intent is HUMINT's territory.

6. Ephemerality. Posts get deleted, accounts get suspended, sites go dark, and links rot. Material that existed when you found it may not exist when someone tries to check your work. Archive at the moment of collection or accept that your finding may become unverifiable.

1.4.3 The Cost Comparison, Concretely

An illustrative comparison for a single mid-complexity question. The figures are ranges for teaching purposes, not quotes.

Approach	Rough cost	Time	Confidence achievable alone	Shareable?
Open source research by one analyst	Analyst time, plus \$0 to a few thousand for imagery or records	Days	Moderate	Yes, freely
Commissioned market research	Tens of thousands	Weeks	Moderate to high	Contractually limited
Licensed investigator or specialist firm	Tens of thousands	Weeks	Moderate to high	Limited
Technical collection on owned systems	Existing infrastructure	Hours	High, within scope	Internally
Recruited human source (government context)	Very high, plus risk to persons	Months to years	High on intent	Severely restricted

The point of the table is not that OSINT is best. It is that OSINT is the only row most organizations can actually execute, and that its output is the only row that can be shared without restriction. Those two facts together explain the discipline's rise.

1.4.4 Why It Matters to You Specifically

Three practical claims, aimed at a student deciding how much attention to give this chapter.

It is the discipline you will actually use. Whatever field you enter, if it involves assessing an external situation, you will do this work. Security, journalism, compliance, sales, insurance, public safety, research, due diligence: all of it is OSINT with domain vocabulary on top.

It is the discipline that will be used on you. Your employer's competitors, prospective employers, opposing counsel, and hostile actors all have access to the same public record. Understanding collection is the prerequisite for understanding exposure. Section 1.10 treats this directly.

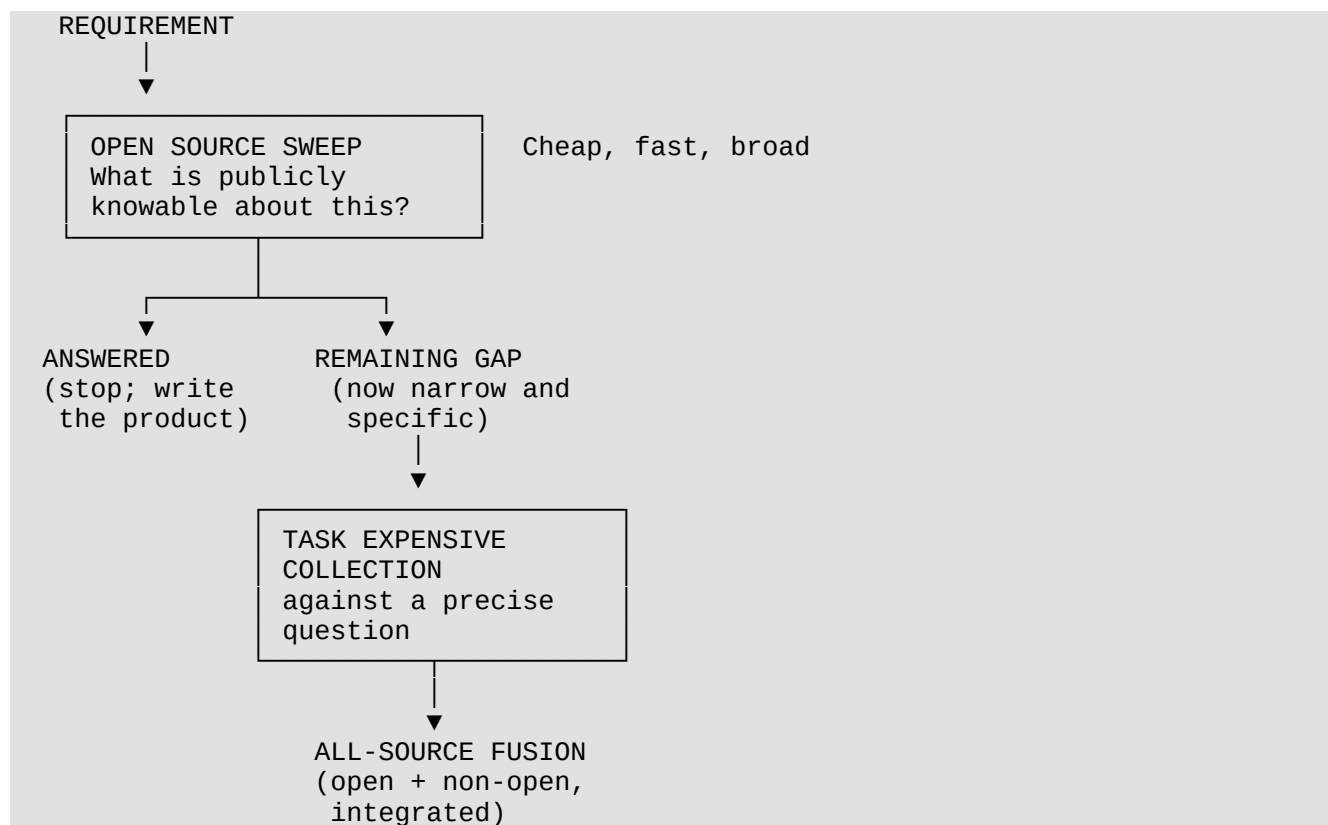
It is where the ethical decisions land. Because the capability is cheap and widely available, there is no institutional gatekeeper deciding whether a given collection is appropriate. You are the gatekeeper. That is a real responsibility and it arrives before you feel ready for it.

1.5 Modern Intelligence and OSINT

1.5.1 From Supporting Role to First Resort

The traditional model treated open sources as background: read the press, then do the real collection. The current model in most mature programs is closer to **open-source-first**: exhaust what is publicly knowable before spending scarce collection capacity, because doing otherwise wastes expensive resources on questions a browser could have answered.

The reordering matters because it changes the sequence of the lifecycle. Under an open-source-first approach, planning and direction is followed by an open source sweep whose purpose is partly to answer the requirement and partly to sharpen it. What survives that sweep becomes the tasking for expensive collection.



The narrowing step is the value. "Tell me about their production capacity" is an expensive tasking. "Confirm whether Line 3 at the Riverside plant is running a second shift" is a cheap one, and open sources are usually what turns the first into the second.

1.5.2 How OSINT Integrates With Other Disciplines

Chapter 1 introduced all-source analysis. OSINT has a specific role in it.

Partner discipline	How OSINT contributes	Example
IMINT / GEOINT	Tells you where and when to image; provides ground-level context for overhead views	A public permit filing identifies the site; commercial imagery shows the construction
HUMINT	Provides the background that makes questions specific, and validates or contradicts source claims	Public filings let you test whether a source's claim about a company is consistent with the record
SIGINT	Provides the identifiers and context that make signals interpretable	Public registration records tie infrastructure to an operator
FININT	Supplies the corporate structure and relationship map around a transaction	Registry filings reveal common directors across shell entities
CTI	Supplies infrastructure enrichment, actor context, and victim identification	Certificate logs and passive DNS expose related attacker domains
MASINT	Provides the ambient context for interpreting a signature	Public weather, seismic, and environmental data explain an anomalous reading

The general pattern: OSINT establishes the frame, and other disciplines fill in what the frame cannot reach. Reverse the order and you spend expensive collection discovering what was in a public filing.

1.5.3 What Modern Practice Looks Like

Four features distinguish current professional practice from amateur research.

Requirements-driven collection. A written requirement, decomposed into observable indicators, before any searching. Without this, OSINT becomes browsing with a deadline.

Systematic documentation. Every finding recorded with its source, URL, access date and time, and an archived copy. Non-negotiable, because links rot and because contested findings must be defensible.

Structured verification. Provenance, corroboration, and internal consistency checked as separate steps rather than absorbed into a general feeling of confidence.

Explicit confidence and gaps. Stated in every product, using a published scale, with the reason for the level given.

Operational security for the researcher. Your collection can be observed. Visiting a small company's careers page repeatedly from a corporate network is itself a signal to them. Professional practice includes managing your own footprint, using research accounts that do not misrepresent identity, and

understanding that some collection is detectable by the subject.

1.5.4 The Three Current Challenges

Challenge 1: synthetic content. Generative models make convincing fake text, images, audio, and video cheap. Two consequences, and the second is worse than the first. The obvious consequence is that fabricated evidence may be believed. The subtler consequence is the **liar's dividend**: once plausible fakes are common, authentic evidence can be dismissed as fake, which benefits anyone who wants a documented event disputed. Verification therefore has to shift from "does this look real" toward provenance: where did this file come from, who first published it, does it appear in an archive predating the claim, and does it corroborate against independent physical evidence such as satellite imagery or weather records.

Challenge 2: commercially available information on individuals. Data brokers aggregate location, purchase, and identity data at scale. Much of it is lawfully purchasable. This creates a genuine policy problem: information that would require legal authorization to collect directly can sometimes be bought. Several jurisdictions have moved to regulate this, and organizations increasingly maintain internal policy limiting purchase of such data even where it is legal. Students should treat the purchase of bulk personal data as outside the scope of this course entirely.

Challenge 3: platform closure and access volatility. Many platforms have restricted programmatic access, changed terms, or removed research tooling. Techniques that worked last year may be unavailable now. The durable skills are therefore method-level rather than tool-level: how to establish provenance, how to corroborate, how to geolocate. Learn the method and treat every specific tool as temporary. This is why no lab in this chapter depends on a particular platform or product.

1.6 Intelligence Failures

1.6.1 How to Read a Failure

Intelligence failures are studied in every serious curriculum, and studied badly in most of them. The bad version treats them as stories about people who were foolish. The useful version identifies which phase of the lifecycle failed, because the remedies are entirely different.

Five distinct failure modes, and one of the most valuable analytic habits you can build is naming which one you are looking at:

Failure mode	What went wrong	Remedy
Collection failure	The information was not gathered	Better sources, new access, different discipline
Processing failure	The information was gathered but never made usable	Translation, normalization, entity resolution, indexing
Analytic failure	The information was available and correctly processed, but the wrong conclusion was drawn	Structured techniques, alternative hypotheses, red teams

Failure mode	What went wrong	Remedy
Dissemination failure	The right conclusion was reached but never reached the decision maker in usable form	Format, channel, timing, sharing rules
Consumer failure	The correct product arrived and was disregarded	Relationship, framing, credibility, governance

Most famous failures are combinations, and most are dominated by the last three rather than the first. That finding is consistent enough across post-mortems that it should change how you think about improvement: the instinct to fix a failure by collecting more is usually wrong.

1.6.2 Failure Case: Pearl Harbor, 1941

What happened. The United States was surprised by the attack on Pearl Harbor despite holding a substantial body of relevant information, including diplomatic decrypts indicating deteriorating relations and imminent action.

Dominant failure modes. Processing, analysis, and dissemination. Material existed across multiple organizations in volumes that outstripped the capacity to translate, collate, and distribute it. Warning signals were present but distributed across separate streams that no single body integrated. Expectations about where an attack would occur shaped how ambiguous indicators were read.

The OSINT angle. Open material contributed to the general picture of deteriorating relations, and this case is the canonical illustration that having the information is not the same as knowing it. The specific lesson relevant to this chapter: **a warning that is never integrated with other warnings is not a warning.** Fragmented indicators, individually explainable, are the normal texture of a pre-event environment.

What better analysis would have changed. A single body responsible for fusing all streams, an explicit indicators-and-warnings framework defining in advance what a surprise attack would look like, and a standing requirement to test the assumption that a distant target was implausible.

1.6.3 Failure Case: The 2002 Iraq WMD Assessment

What happened. Pre-war assessments concluded that Iraq maintained active weapons of mass destruction programs. Post-war investigation found the programs were not as assessed. Subsequent official reviews examined the analytic process in detail.

Dominant failure modes. Analysis, overwhelmingly, with a contributing collection failure.

The reviews identified a cluster of problems that together form the best-documented case of analytic failure available for study:

- **Assumption treated as fact.** The premise that programs continued was treated as established rather than as a hypothesis requiring support, and evidence was then interpreted through it.
- **Layering.** Judgments from earlier products were carried forward as established background, so

uncertainty disappeared across successive documents without anyone deciding to remove it.

- **Source reliability failures.** Reporting from sources with clear incentives to mislead, or with unverified access, was weighted more heavily than their evaluation warranted.
- **Absence of alternatives.** The hypothesis that programs had been abandoned was not developed with comparable rigor.
- **Confidence language misuse.** The strength of the language conveyed to consumers exceeded the strength of the underlying evidence.
- **Dissent not surfaced.** Where technical disagreement existed, it was not prominent enough in the summary judgments that decision makers actually read.

The OSINT angle. International inspection reporting, technical publications, and public analysis contained material inconsistent with parts of the assessment. The lesson is not that open sources had the answer. It is that open sources contained the disconfirming material, and a process weighted toward confirming an existing premise had no mechanism to give that material weight.

What better analysis would have changed. Analysis of Competing Hypotheses with abandonment developed as a genuine alternative; a key assumptions check on the central premise; source reliability re-evaluation independent of how well the reporting fit; explicit separation of what was known from what was inferred; and dissent carried into the summary rather than into footnotes.

Why this case is the most important one in this chapter. Nearly every failure mode taught in Chapter 1 appears in it, and none of them involve a shortage of information.

1.6.4 Failure Case: The 2013 Crowdsourced Misidentification

What happened. Following a mass-casualty bombing at a public sporting event in the United States, users on social platforms and discussion forums attempted to identify suspects from publicly posted photographs of the crowd. The effort misidentified multiple innocent people. One misidentified individual was a missing student who had died before the event, and his family was subjected to public accusation during their search for him. Some media outlets amplified the false identifications before official information was released.

Dominant failure modes. Analysis and dissemination, with a specific structural cause: no accountability and no verification threshold.

The mechanics of the failure are worth stating precisely, because they recur constantly at smaller scale:

- **Pattern matching mistaken for identification.** A person carrying a backpack in a crowd photograph matches thousands of people. Resemblance is not identification.
- **Confirmation cascade.** Once a candidate was proposed, participants sought features that fit and discounted features that did not.
- **No confidence discipline.** Assertions circulated without any expression of uncertainty, and the language hardened as it spread.
- **Irreversible dissemination.** Publication at internet scale cannot be retracted. The correction

never reaches the same audience as the accusation.

- **Diffused responsibility.** No individual owned the judgment, so no individual bore the cost of it being wrong.

Why this case belongs in an OSINT chapter. It is the discipline's clearest cautionary tale, and it is not a story about bad people. Many participants intended to help. The techniques used were recognizably OSINT techniques. What was missing was every professional control: a requirement, a verification standard, a confidence statement, and an accountable analyst. This is what OSINT looks like without tradecraft, and it is why Rule 1 at the front of this chapter exists.

What better practice would have changed. A verification threshold requiring corroboration from an independent source before naming anyone; a rule against publishing identifications of private individuals at all, on the grounds that the potential harm is severe and the public benefit is served by passing material to investigators rather than publishing it; and explicit confidence language that would have exposed how thin the basis was.

1.6.5 Failure Case: Missed Warning in a Corporate Setting

Composite case, drawn from documented patterns in post-incident reviews.

What happened. A manufacturer relied on a single supplier for a specialized component. The supplier's home region experienced a natural disaster, halting production for eleven weeks and costing the manufacturer a significant share of annual operating profit.

What was publicly knowable beforehand. The supplier's facility location was in public filings. The region's hazard exposure was in public government hazard mapping. The supplier's own annual report disclosed concentration of production at that single site. Trade press had covered the region's exposure. Every element of the risk was in open sources, and none of it was assembled.

Dominant failure modes. Planning and direction, first and foremost. Nobody had a requirement that read "identify single points of failure in our supply base and their geographic hazard exposure." Without the requirement, the collection never happened, even though the collection would have been trivial.

Why this is the most common failure pattern in commercial settings. Corporate intelligence failures are rarely dramatic. They are usually a public fact that nobody was assigned to look for. The remedy is unglamorous: standing requirements, reviewed on a schedule.

1.6.6 The Pattern Across Failures

Case	Dominant failure mode	Was the information available?	Cheapest available remedy
Pearl Harbor	Processing, analysis, dissemination	Largely yes	Fusion of streams; indicators framework
Iraq WMD 2002	Analysis	Partly, including disconfirming material	Competing hypotheses; assumptions check

Case	Dominant failure mode	Was the information available?	Cheapest available remedy
2013 misidentification	Analysis, dissemination	Yes, and it was misread	Verification threshold; publication restraint
Supply chain (composite)	Planning and direction	Entirely yes	A written standing requirement

Look at the third column. In every case the information was substantially available. This is the central empirical finding of failure studies, and it is the reason this chapter spends more pages on verification and analysis than on collection technique.

1.7 Intelligence Successes

1.7.1 Success Case: Aircraft Shoot-Down Reconstruction, 2014 Onward

The situation. A civilian airliner was shot down over an active conflict zone in eastern Ukraine in July 2014, killing all aboard. Responsibility was contested, with competing state narratives.

The investigative process. Independent open source investigators, working alongside and sometimes ahead of official inquiries, reconstructed the movement of a mobile surface-to-air missile system using exclusively public material:

Step	Method	Material used
1	Collect participant imagery	Social media photographs and videos of a missile transporter posted by residents along its route
2	Geolocate each image	Match visible buildings, road markings, signage, power lines, and terrain against street-level imagery and satellite views
3	Chronolocate	Shadow angles, weather conditions, and post timestamps to establish sequence
4	Identify the specific vehicle	Distinctive physical features of the individual launcher visible across multiple images, allowing one vehicle to be tracked rather than a vehicle type
5	Reconstruct the route	Assemble the geolocated, time-ordered sightings into a continuous path

Step	Method	Material used
6	Attribute the unit	Cross-reference against publicly posted material from military personnel, unit insignia, and convoy imagery from the originating region
7	Publish with sourcing	Release findings with every source cited so that the reconstruction could be independently checked

Outcome. The open source reconstruction was substantially consistent with the conclusions of subsequent official investigations, and the methodology was cited in later legal and diplomatic proceedings. The broader outcome mattered more than the specific finding: it demonstrated that rigorously documented open source work could withstand adversarial scrutiny from a state actor with every incentive to discredit it.

Why it worked. Four things, all of which you can replicate.

1. **Physical evidence anchored the claims.** Buildings and roads do not move. Geolocation against fixed features is checkable by anyone.
2. **Independence of sources.** Dozens of unconnected residents posting from different locations cannot easily be coordinated into a false picture.
3. **Total sourcing transparency.** Every claim was traceable, which made the work reproducible and therefore defensible.
4. **Specificity.** Identifying an individual vehicle rather than a vehicle type raised the evidentiary weight enormously.

1.7.2 Success Case: Identification of State Operatives, 2018

The situation. Two men were identified by British authorities as suspects in a nerve agent poisoning in Salisbury, England, using cover identities. The relevant government denied any connection.

The investigative process. Open source investigators, working with journalists, established the men's actual identities and institutional affiliation using:

- Published photographs of the suspects released by authorities.
- Facial comparison against images in publicly circulating and leaked-then-public document sets.
- Analysis of identity document numbering patterns, where sequential or anomalous issuance revealed administrative handling inconsistent with ordinary civilians.
- Public registry and address records, and publicly available databases circulating in the region.
- Cross-referencing of travel records that had become public.

- Confirmation through public honors and award records.

Outcome. Real names and institutional affiliations were published and were not credibly refuted. The findings reframed the diplomatic dispute, because a denial is harder to sustain against a named and documented individual than against an unnamed allegation.

Why it worked. Administrative systems leave patterns. Cover identities have to be created inside real bureaucracies, and bureaucracies produce records with structure. The transferable lesson: **anomalies in administrative data are often more revealing than the data itself.** A document number that does not fit the expected distribution is a finding.

The ethical note this case requires. The subjects were state operatives implicated in an attempted killing, identified by journalists working in the public interest with careful verification. The methods used are not thereby licensed for use against private individuals. The public interest justification does real work here, and it does not transfer to someone researching a neighbor. Test 4 in Section 1.1.4 applies.

1.7.3 Success Case: Public Pre-emptive Disclosure, 2021 to 2022

The situation. In the months before the full-scale invasion of Ukraine in February 2022, military buildup was extensively documented in open sources, including commercial satellite imagery, publicly posted video of rail and road movements of military equipment, and geolocation of unit deployments.

What was different about this case. Two things, and both were novel enough to change practice.

First, **civilian analysts produced a running public order-of-battle** from commercial imagery and social video, tracking specific units from home garrisons to staging areas. This was work that had historically required national reconnaissance capability.

Second, **governments deliberately disclosed assessments publicly**, in advance, in unusual detail. The reasoning was partly that pre-emptive disclosure narrows an adversary's options for a false pretext, and partly that open source corroboration meant the disclosures could be independently checked by journalists and analysts, which made them credible in a way that unsupported official claims are not.

Outcome. The invasion was not prevented. But the sequence of events was widely documented and independently corroborated before it occurred, which substantially constrained the space for competing narratives afterward, and allied coordination and preparation proceeded on a shared factual basis.

Why it matters for this chapter. It is the clearest demonstration of OSINT's shareability advantage from Section 1.4.1. A classified assessment of the buildup would have been more detailed and less useful for that specific purpose, because it could not have been shown to the public and checked.

1.7.4 Success Case: Large-Scale Explosion Reconstruction, 2020

The situation. A catastrophic explosion at the port of Beirut in August 2020 killed and injured thousands and devastated a large area of the city. The cause and the chain of responsibility were immediately contested.

The investigative process. Within days, open source analysts and newsroom investigation teams:

- Collected and geolocated dozens of civilian videos capturing the event from different angles and distances.
- Used the time offset between the visible flash and the audible arrival of the blast in each video, combined with known geolocation, to estimate positions and yield.
- Established the pre-event condition of the warehouse from satellite imagery archives and prior public reporting.
- Traced the cargo's origin through public shipping records, vessel tracking history, and court filings from the vessel's abandonment.
- Recovered years of public correspondence and official documents showing that the hazard had been repeatedly flagged to authorities.

Outcome. A detailed public reconstruction of both the physical event and the administrative history preceding it, available within days to weeks rather than the months or years an official inquiry requires. It informed public accountability efforts and press coverage, and it stands as a demonstration that documentary and physical open source work combine well.

Why it worked. Physics is checkable. The speed of sound does not vary with politics, so the blast-timing method produced a result that could be reproduced by anyone with the same videos. Combining that with a documentary trail from public records made the finding hard to dismiss on either physical or bureaucratic grounds.

1.7.5 What the Successes Have in Common

Feature	Present in all four cases
Anchoring in physical or documentary fact	Buildings, road features, shipping records, document numbering, blast physics
Multiple genuinely independent sources	Many unconnected publishers, or independent record systems
Total sourcing transparency	Every claim traceable and reproducible by a skeptic
Specificity over generality	Individual vehicles, named individuals, specific units, specific cargo
Method robustness under adversarial scrutiny	All four were attacked by interested parties and survived
Ethical grounding	Public interest subject matter, not private individuals

Compare that to the failure cases. The successes are not distinguished by better access. They are distinguished by discipline. Every technique in the success column was available to the participants in the 2013 misidentification, and the difference in outcome came from verification thresholds and accountability, not from tools.

1.8 Government Use of OSINT

1.8.1 The Range of Government Applications

"Government" is not one consumer. At minimum it is six, with different requirements and different legal authorities.

Government function	Typical OSINT applications
National security and foreign policy	Foreign leadership statements and media analysis, order-of-battle tracking, sanctions evasion monitoring, treaty compliance indicators, political stability assessment
Military	Terrain and infrastructure assessment, target-area context, adversary doctrine from published sources, force disposition from public imagery and social posting
Law enforcement	Investigative lead development, threat-to-life assessment, fugitive location through public records, organized crime network mapping, event security planning
Regulatory and enforcement agencies	Unlicensed activity detection, false advertising and consumer protection monitoring, environmental compliance from imagery, financial fraud indicators
Emergency management and public health	Damage assessment from imagery and civilian posting, situational awareness during events, outbreak signal detection from public reporting, resource needs identification
Diplomatic and public affairs	Foreign information operation detection, public sentiment tracking, counter-disinformation, attribution support for public statements

1.8.2 What Governments Can Do That Others Cannot

Not more searching. Different things:

- **Purchase at scale.** Comprehensive commercial imagery subscriptions and large data licenses.
- **Sustain longitudinal coverage.** Monitoring a country's media continuously for decades, including languages no commercial entity would fund.
- **Fuse with non-open sources.** Producing validated OSINT by corroborating against classified reporting.
- **Act on findings with legal authority.** A government can convert an OSINT lead into a

warrant, a subpoena, a sanction, or a diplomatic action. That conversion pathway is the real difference.

- **Employ language and regional depth.** Analysts with deep expertise in narrow areas that no market would support.

1.8.3 The Constraints Governments Face That Private Actors Do Not

This is the part students find counterintuitive. In several respects, governments are *more* restricted in open source collection than companies are.

- **Constitutional and statutory limits on collecting about their own population.** In many democracies, agencies face specific restrictions on collecting and retaining information about citizens and residents, even from public sources. The public nature of the source does not remove the restriction.
- **Criminal predicate and purpose limitation.** As covered in Chapter 1, retaining records about individuals typically requires a defined predicate and purpose, with retention limits and audit trails.
- **First Amendment and equivalent protections.** Collecting on lawful political, religious, or associational activity is restricted, and monitoring protest activity has been a recurring subject of litigation and policy revision.
- **Oversight and disclosure.** Inspectors general, legislative committees, courts, and transparency laws mean government open source programs are auditable in ways corporate programs are not.
- **Policy limits on purchased data.** Recognizing the problem in Section 1.5.4, governments have increasingly adopted internal policy, and in some cases legislation, restricting acquisition of commercially available personal data that would otherwise require legal process to obtain.

The teaching point. "It is public" answers a legal question about the source. It does not answer the separate legal question about the collector's authority. A student who understands that distinction is ahead of most practitioners.

1.8.4 Government OSINT in Practice: An Illustration

A state fusion center receives a report of concerning statements on a public forum ahead of a large public event.

Phase	Activity	Constraint applied
Planning	Requirement defined narrowly around a specific threat-to-life question, with a stated deadline	Requirement documented to establish purpose
Collection	Publicly viewable posts collected; no false accounts created; no private groups accessed	Terms of service and identity honesty observed

Phase	Activity	Constraint applied
Processing	Posts preserved with timestamps and archived; account handles resolved carefully to avoid conflating similar names	Documentation for later legal defensibility
Analysis	Assessment of whether statements constitute a specific, credible threat, distinguishing protected speech from threat indicators	Speech protections applied explicitly
Dissemination	Product shared with event security and the relevant agency, with handling caveats	Need to know; retention decision recorded
Feedback	Post-event review; records purged or retained per the retention schedule based on whether predicate was established	Purpose limitation and retention limits

Notice that the constraints appear in every phase, not as a compliance step at the end. That is what mature practice looks like.

1.9 Commercial Use of OSINT

1.9.1 The Commercial Application Map

Business function	OSINT applications
Competitive intelligence	Competitor capability, expansion signals, pricing behavior, hiring patterns, product roadmap inference
Cyber threat intelligence	Threat actor tracking, exposed asset discovery, credential leak monitoring, brand impersonation detection, supply chain exposure
Third-party and vendor risk	Ownership and beneficial ownership research, litigation and sanctions screening, facility hazard exposure, financial distress indicators
Mergers, acquisitions, and investment	Target diligence, undisclosed litigation, key-person risk, market position validation, on-the-ground verification of claims
Physical and executive security	Site threat assessment, travel risk, event security, protest and disruption monitoring, executive digital exposure reduction

Business function	OSINT applications
Insurance and underwriting	Property and catastrophe exposure from imagery, claims verification, risk scoring
Fraud and financial crime	Shell company networks, synthetic identity indicators, merchant fraud patterns, counterfeit and diversion detection
Brand, legal, and IP protection	Counterfeit marketplaces, trademark infringement, gray market diversion, unauthorized resellers
Supply chain resilience	Supplier concentration, sub-tier discovery, port and shipping disruption, single points of failure
Recruiting and market research	Talent market mapping, customer sentiment, market entry conditions

1.9.2 The Legal Frame for Commercial OSINT

Companies do not have government authorities, and they have obligations governments do not. Six areas to know:

1. Privacy and data protection law. Regimes such as the EU General Data Protection Regulation and various United States state statutes govern the processing of personal data, and the public availability of data does not automatically permit its collection, storage, or use. Where such regimes apply, an organization typically needs a lawful basis, a defined purpose, a retention limit, and in some cases a duty to inform the individual. This surprises people: scraping publicly posted personal data can be a regulated processing activity.

2. Computer misuse law. Statutes such as the Computer Fraud and Abuse Act in the United States and equivalents elsewhere prohibit unauthorized access. Credential use, control circumvention, and access after revocation are the danger zones. The law in this area continues to develop, particularly around scraping of public pages, and organizations should not treat unsettled law as permission.

3. Terms of service and contract. Platform terms may prohibit automated collection, account creation for research, or use of data for particular purposes. Breach is typically a contract matter rather than a crime, but it can be both, and it can void the admissibility or usability of findings.

4. Trade secret law. Carried forward from Chapter 1: acquiring information whose owner took reasonable measures to protect it can be misappropriation. OSINT stays clear of this by definition, and drifts into it when researchers accept material from insiders.

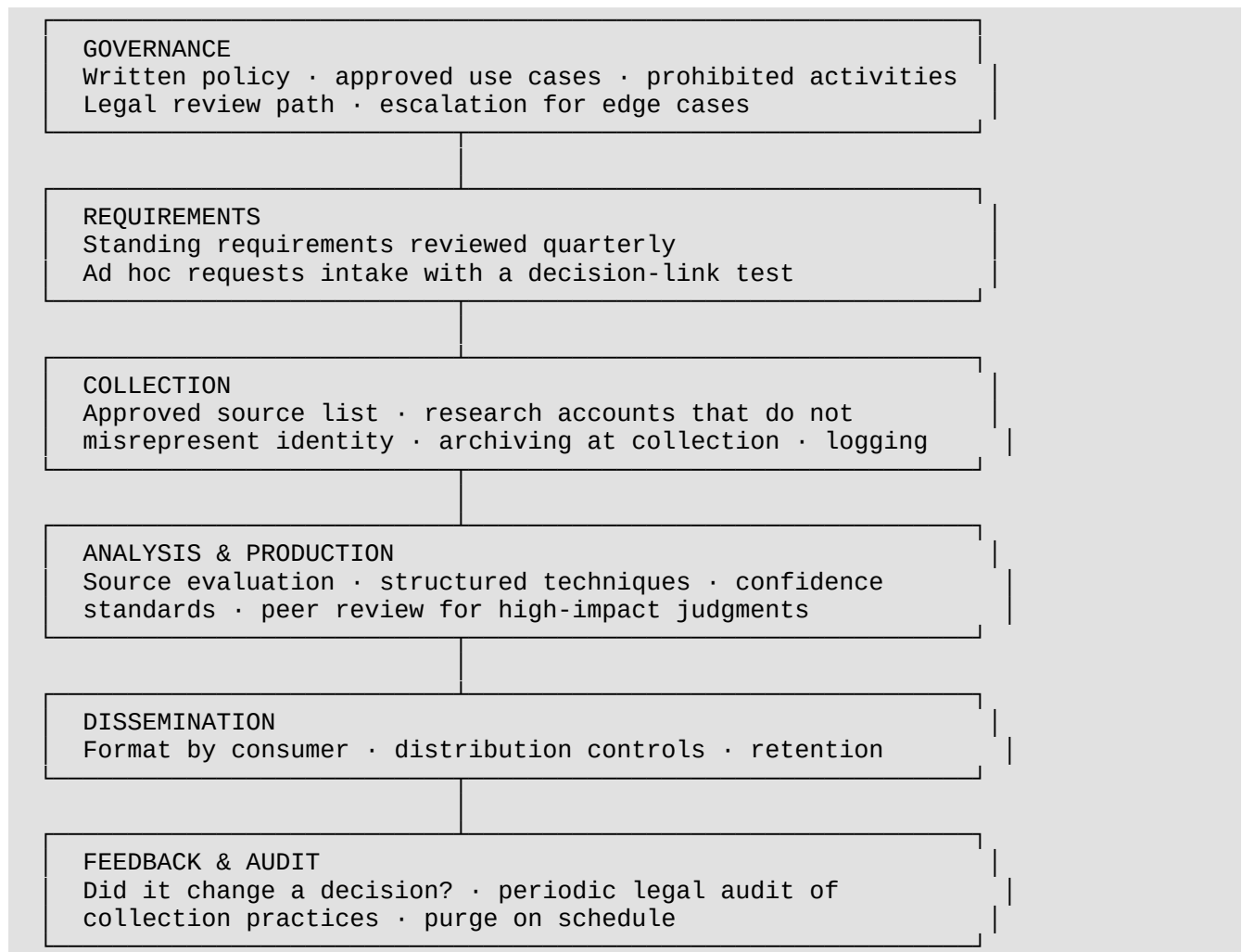
5. Fair Credit Reporting Act and equivalents. In the United States, using information to make decisions about employment, credit, insurance, or housing may bring an activity within consumer reporting regulation, with attendant notice, accuracy, and dispute obligations. A hiring manager doing informal social media research on a candidate may be creating regulatory exposure they do not know about.

6. Sector-specific rules. Financial institutions, healthcare organizations, and regulated utilities all have

additional obligations governing information handling.

1.9.3 A Commercial OSINT Program, Structured

What a functioning corporate capability actually contains:



The two boxes most often missing in real programs are Governance at the top and Feedback and Audit at the bottom. A program without governance eventually does something indefensible. A program without feedback cannot demonstrate value and eventually loses funding.

1.9.4 The Prohibited List Every Commercial Program Should Have

Written in advance, so the decision is not made under deadline pressure:

- No creation of accounts using false identities to access restricted content.
- No access to any system, account, or file requiring credentials the organization was not granted.
- No engagement with insiders at other organizations regarding confidential matters.
- No purchase of bulk personal data about individuals.

- No collection about individuals' protected characteristics, political or religious activity, or health.
- No collection about family members, including minors, of any subject.
- No investigation of employees outside a documented HR or legal process.
- No use of findings in hiring, credit, insurance, or housing decisions without routing through the compliant process.
- No collection that would be indefensible if published, including if published by the subject.

That last item is the practical test, and it generalizes well: **would you be comfortable if the subject read your collection log?** If not, stop before you collect, not after.

1.10 Personal Use of OSINT

1.10.1 Legitimate Personal Applications

OSINT skills have real defensive and practical value for individuals. The legitimate uses divide into four categories.

1. Self-assessment: knowing your own exposure. Search for yourself deliberately and systematically: name variants, old usernames, email addresses in breach-notification services, public records, property records, photographs, and information about you posted by others. The purpose is to find what an adversary would find. This is the single most useful application in this section and the one most students should actually perform.

2. Personal safety and due diligence. Verifying that a landlord actually owns the property, that a contractor holds a license and has no pattern of judgments, that a charity is registered, that a job offer comes from a real company at a real address, that a marketplace seller is not running a known pattern. This is verification of a counterparty you are about to transact with, and it is entirely appropriate.

3. Consumer protection and fraud avoidance. Recognizing the structure of common frauds: recently registered domains impersonating brands, review patterns that indicate manipulation, investment offers with unregistered promoters, rental listings copied from real ones. Section 3.7 develops this as a full case study.

4. Civic and research participation. Contributing to public-interest verification work, local journalism, historical research, genealogy, disaster mapping, and similar activity.

1.10.2 The Personal Ethics Line

Personal use is where the discipline's ethics get hardest, because there is no employer, no policy, and no oversight. Only you.

What is not acceptable, regardless of how public the information is:

- Researching a former or current partner, an ex, or someone who has asked you to stop

contacting them. This is the most common misuse of these skills and it can constitute stalking under the law in many jurisdictions regardless of the sources used.

- Researching a person you are in a dispute with, in order to gain leverage.
- Compiling profiles of private individuals for any reason not tied to a specific transaction you are about to enter.
- Publishing identifying information about a private individual. This is doxxing, it causes documented severe harm, and public availability of each component is not a defense.
- Researching minors.
- Investigating a person on someone else's behalf informally, which removes even your own judgment from the loop.

Three questions to ask before any personal research on a person:

1. **Am I about to transact with this person or organization?** If yes, verification proportionate to the transaction is appropriate. If no, be careful.
2. **Would I be comfortable telling them exactly what I searched for and why?** The identity test from Section 1.1.4, applied reflexively.
3. **Am I looking for a specific fact, or am I building a profile?** A specific fact tied to a decision is verification. An open-ended profile of a private person is surveillance, and the mosaic effect means it produces something the person never made public.

1.10.3 Reducing Your Own Exposure

A practical checklist, which doubles as a demonstration of what collection can reach.

Exposure area	What is often findable	Reduction step
Data broker aggregation	Name, address history, relatives, phone, age	Submit removal requests to major aggregators; repeat periodically, since records repopulate
Breach data	Email addresses, reused passwords	Check breach notification services; use unique passwords and a manager; enable multifactor authentication
Social media	Location patterns, workplace, schedule, family, home interior and exterior	Audit privacy settings; remove location metadata; reconsider photographs showing your home, vehicle, or children's schools
Professional profiles	Employer, role, tenure, colleagues, technology stack	Accept this exposure as the cost of professional visibility, but avoid detail that reveals internal systems

Exposure area	What is often findable	Reduction step
Public records	Property ownership, voter registration in some jurisdictions, licenses, court filings	Largely irreducible; know what is exposed rather than expecting to remove it
Photograph metadata	GPS coordinates, device, timestamp	Strip metadata before posting; most platforms strip it, but not all, and not always
Username reuse	Links otherwise separate accounts into one identity	Use distinct usernames for distinct contexts
Family and associates	Others post about you without your involvement	Ask, and accept limited control

That last row is worth sitting with. Much of your exposure is created by other people, and you cannot remove it. The realistic goal of personal OSINT hygiene is not invisibility. It is knowing your exposure well enough to plan around it.

1.10.4 Personal Use as a Learning Path

Because personal use has no institutional oversight, it is also where students build habits. Build good ones now:

- Document even casual research. It creates the reflex.
- State your confidence out loud, even to yourself. "I am fairly sure this is the same company" is different from "this is the same company."
- Notice when you have moved from answering a question to satisfying curiosity about a person. Stop at the transition.
- When you are wrong, note what misled you. A personal error log is the cheapest calibration tool available.

1.11 Verification: The Core Tradecraft

This section is the practical heart of the chapter. Sections 1.6 and 1.7 showed that the difference between failure and success is verification discipline. Here is what that discipline consists of.

1.11.1 The Provenance Chain

Before assessing whether a claim is true, establish where it came from. Five questions, in order:

1. **Who published this first?** Not who you saw it from. The original.
2. **When was it first published?** Compared against when the described event occurred.
3. **Does the publisher have a plausible basis for knowing it?** Presence, access, or expertise.

4. **Has this content appeared before, in another context?** Recycled footage from an older event is the most common form of open source deception.

5. **What is the publisher's interest in the claim being believed?**

Any claim that cannot answer questions 1 and 2 should carry an explicit caveat in your product.

1.11.2 The Verification Techniques

Technique	What it establishes	How it works	Limits
Reverse image search	Whether an image is original or recycled	Query the image against indexed collections to find earlier appearances	Cropping, mirroring, and re-encoding defeat naive matching; check multiple services and try cropped regions
Geolocation	Where a photo or video was taken	Match fixed features (buildings, signage, road markings, terrain, vegetation, utility infrastructure) against street-level and satellite imagery	Requires distinctive features; hard in uniform terrain
Chronolocation	When it was taken	Shadow direction and length against sun position for the location; weather records; vegetation state; visible clocks or dated objects	Requires a known location first; overcast conditions eliminate shadow analysis
Metadata examination	Device, time, and sometimes coordinates	Read embedded file metadata where present	Most platforms strip metadata on upload; metadata can be edited, so it is corroborating rather than conclusive
Cross-platform correlation	Whether multiple independent publishers captured the same event	Locate the event from different accounts, angles, and platforms	Coordinated posting can mimic independence; check account histories
Account history analysis	Whether a publisher is authentic	Account age, posting history consistency, network, language patterns, activity pattern	Aged accounts can be purchased or compromised
Satellite comparison	Whether physical change occurred and when	Compare archived imagery before and after the claimed date	Revisit intervals limit temporal precision; clouds obscure
Documentary corroboration	Whether records support the claim	Registries, filings, dockets, permits, procurement notices	Records lag reality; absence proves little

Technique	What it establishes	How it works	Limits
Physical consistency	Whether the content obeys physics	Sound delay, shadow geometry, reflections, scale relationships, weather consistency	Requires domain knowledge; strongest check available when applicable

1.11.3 Detecting Synthetic and Manipulated Content

Given Section 1.5.4, the practical guidance:

Deprioritize artifact hunting. Looking for visual tells (hands, teeth, background warping) is unreliable and getting worse as generation quality improves. Do not build your assessment on it.

Prioritize provenance and corroboration. A video of an event that actually happened will usually have companions: other angles, other publishers, weather that matches records, physical features that match satellite imagery, and a first appearance whose timing makes sense. Synthetic content typically fails at least one of these, not because the pixels are wrong but because the surrounding evidentiary context does not exist.

Treat single-source, single-angle, high-emotional-impact content as unverified by default.

Especially when it arrives fast, aligns perfectly with a contested narrative, and cannot be traced to an original publisher.

Name the uncertainty in the product. "Content consistent with the claim, but unverified; no corroborating angle located, first appearance cannot be established" is a professional statement. Silence about it is not.

1.11.4 Recognizing Coordinated Inauthentic Behavior

Signals that a body of open source material has been manufactured rather than observed:

- Accounts created in tight clusters, often with sequential or template-generated names.
- Near-identical phrasing across accounts claiming to be independent.
- Posting activity concentrated in hours inconsistent with the claimed location.
- Engagement patterns without organic conversation: high amplification, no replies of substance.
- Content that appears first on a low-credibility outlet and then propagates to accounts claiming firsthand knowledge.
- A claim that is highly specific about the conclusion and vague about the observable details.

The analytic response is not to dismiss the topic. It is to note that the material cannot support a judgment and to state what would.

1.11.5 The Documentation Standard

Every finding, without exception:

Field	Why
Claim, in one sentence	Forces precision about what is actually being asserted
Source name and URL	Reproducibility
Date and time of access, with time zone	Because content changes and disappears
Archived copy	Because links rot; archive at the moment of collection
Publisher type and apparent independence	Circular reporting detection
Reliability rating (A to F)	Chapter 1 framework
Credibility rating (1 to 6)	Chapter 1 framework
Verification steps performed	So a reviewer can see what was and was not checked
Analyst initials	Accountability

An OSINT product without this trail is an opinion. With it, it is evidence someone else can test. This is graded in every lab in this chapter.

Part 2: Comparison Tables

2.1 Information vs. OSINT vs. Intelligence

Dimension	Open source information (OSIF)	Open source intelligence (OSINT)	Intelligence (all-source)
What it is	Public material collected and given basic context	OSIF evaluated and analyzed against a requirement, with a judgment	A judgment fused from all available disciplines
Driver	Availability or general interest	A stated requirement	A stated requirement
Source range	Public only	Public only	Public plus human, technical, imagery, financial, and others
Evaluation	Often none	Reliability and credibility assessed	Reliability and credibility assessed across disciplines
Judgment	None	Explicit, with confidence level	Explicit, with confidence level
Who can produce it	Anyone	Anyone with tradecraft	Organizations with multi-discipline access

Dimension	Open source information (OSIF)	Open source intelligence (OSINT)	Intelligence (all-source)
Shareability	Full	Full	Restricted by classification and source protection
Auditability by a third party	Full	Full	Limited
Reaches intent?	Rarely	Rarely	Sometimes, via human sources
Typical failure	Compilation mistaken for analysis	Unverified material treated as established	Over-weighting a single privileged source
Cost	Low	Low	High
Example	A folder of a company's filings and news coverage	"The company will likely open a Southeast facility in Q4; moderate confidence"	The same judgment, corroborated by a source with internal access

2.2 Traditional (Closed-Source) Intelligence vs. OSINT

Dimension	Traditional closed-source intelligence	OSINT
Access requirement	Clearance, authority, and often statutory basis	None beyond lawful public access
Cost per finding	High to very high	Low
Time to first result	Days to months	Minutes to days
Risk to persons	Can be severe (sources, collectors)	Minimal, though researcher OPSEC still matters
Volume available	Constrained by collection capacity	Effectively unlimited; triage is the constraint
Deception exposure	Present, but adversary must penetrate a protected channel	High; publishing is the cheapest route to an analyst
Verification burden	Lower per item; sourcing is documented internally	High per item; provenance must be established
Shareability of findings	Restricted; may be unusable in public or legal settings	Full; usable in court, press, and public disclosure
Reproducibility by a skeptic	Not possible	Fully possible
Reaches intent and deliberation	Yes, uniquely	Rarely
Coverage of denied areas	Yes, this is its purpose	Poor; open reporting is thin where publishing is dangerous

Dimension	Traditional closed-source intelligence	OSINT
Coverage of ordinary commercial and civic questions	Poor; not built for it	Strong
Institutional prestige historically	High	Low, now rising
Dominant modern constraint	Access and risk	Verification and volume
Best used for	Intent, denied areas, protected plans	Framing, tipping and cueing, breadth, anything shareable

2.3 Government vs. Commercial vs. Personal OSINT

Dimension	Government	Commercial	Personal
Primary purpose	National security, public safety, regulation	Profit, risk reduction, competitive position	Self-protection, transaction due diligence, civic participation
Typical consumer	Policymakers, commanders, investigators	Executives, security teams, legal, deal teams	Yourself
Budget	Large, sustained	Modest to substantial	Essentially none
Tooling	Enterprise, plus bespoke	Commercial platforms and subscriptions	Free tools and a browser
Data purchase capability	Extensive, but increasingly policy-restricted	Extensive, subject to privacy law	Minimal, and should be avoided
Legal authority to act on findings	Warrant, subpoena, sanction, enforcement	Contract, litigation, referral to authorities	Decline the transaction; report to authorities
Restrictions on collecting about individuals	Strictest: constitutional limits, predicate, retention, oversight	Moderate: privacy law, FCRA-type rules, contract	Least formal, and therefore the highest personal responsibility
Oversight	Inspectors general, legislatures, courts, transparency law	Internal legal, audit, regulators	None; only your judgment
Documentation expectation	Formal, auditable, retention-scheduled	Formal for legal defensibility	Usually none, which is a habit worth breaking
Sharing pathway	Interagency, allied, and increasingly public disclosure	Sector sharing groups, vendors, customers	Rarely appropriate to share at all
Characteristic failure	Over-collection on protected activity; poor	Requirement never written, so obvious risks	Drifting from verification into

Dimension	Government	Commercial	Personal
	integration	are missed	surveillance of a person
Skill emphasis	Language, region, longitudinal analysis	Corporate records, technical exposure, financial structure	Self-exposure assessment, counterparty verification, fraud recognition

2.4 Intelligence Successes vs. Intelligence Failures

Dimension	Successes (Section 1.7)	Failures (Section 1.6)
Requirement	Specific and answerable	Absent, vague, or never written
Anchoring evidence	Physical or documentary, checkable by anyone	Inference chains resting on assumption
Source independence	Many genuinely unconnected publishers or record systems	Apparent corroboration tracing to one origin
Treatment of alternatives	Competing explanations developed and tested	Single explanation adopted early and defended
Assumptions	Written down and tested	Treated as established fact
Confidence language	Calibrated to the evidence, stated explicitly	Stronger than the evidence supported, or absent
Sourcing transparency	Total; work reproducible	Opaque; claims not traceable
Accountability	A named analyst or team owns the judgment	Diffuse or absent
Dissemination	Reached the consumer in usable form and time	Fragmented, delayed, buried, or irreversible
Disconfirming evidence	Sought and weighted	Discounted, or never looked for
Handling of individuals	Public-interest subjects, verification thresholds	Private individuals named without corroboration
Response to scrutiny	Survived adversarial challenge	Collapsed under review
Was information available?	Yes	Also yes, in nearly every case

That last row is the row that matters. Availability of information does not distinguish success from failure. Discipline does.

2.5 Advantages and Limitations of OSINT

Advantage	Why it matters	Corresponding limitation	Mitigation
Low cost	Puts capability within reach of small teams and individuals	Cheapness encourages aimless collection	Written requirements before collection
Speed	Often the only source in the first hours of an event	Fast conclusions are frequently wrong	Verification threshold before any judgment
No clearance required	Wide analyst and consumer pool	Wide pool includes untrained participants	Tradecraft training; explicit confidence discipline
Fully shareable	Usable in court, press, public disclosure, partner sharing	Publication is irreversible and can harm the innocent	Restraint rules for private individuals; corroboration before naming
Reproducible by third parties	Findings defensible under adversarial scrutiny	Requires disciplined documentation to realize	Mandatory collection log and archiving
Enormous breadth	Covers questions no other discipline addresses	Uneven coverage; biased toward connected populations	Treat absence of reporting as weak evidence; state coverage gaps
Tips and cues expensive collection	Multiplies the value of scarce resources	Can create false confidence that the picture is complete	Explicitly state what open sources cannot reach
Adversarially peer-reviewed community	Errors get caught publicly	Public work exposes methods to adversaries	Accept the trade; it is generally favorable
No risk to human sources	Ethically and operationally simpler	Rarely reaches intent or private deliberation	Pair with other disciplines where intent matters
Auditable and legally usable	Supports accountability and prosecution	Subject to privacy, computer misuse, and consumer reporting law	Written policy, legal review, prohibited-activity list

Part 3: Case Studies

Each case follows the same structure: the situation, the requirement, what was collected, the investigative process, the outcome, and the transferable lesson. Cases marked *historical* describe real documented events. Cases marked *composite* are built from documented patterns and are not real

organizations; never cite them as real.

3.1 National Security: Tracking Sanctions Evasion at Sea

Historical pattern, described in general terms

The situation. Following the imposition of sanctions and price caps on certain petroleum exports after 2022, a large volume of maritime trade shifted toward vessels operating outside mainstream insurance and flag arrangements, commonly described in trade press as a shadow or dark fleet. Governments, insurers, and journalists needed to understand which vessels were involved, how cargo origin was being obscured, and which service providers were enabling it.

The requirement. Which vessels are transporting sanctioned cargo, by what method, and which intermediaries in the shipping, insurance, and flag registry chain are facilitating it? The decision it supported: enforcement targeting, insurance underwriting, and port state control priorities.

What was collected.

Source	Material
AIS vessel tracking	Position, speed, heading, and identity broadcasts from ships, publicly available through commercial and free platforms
Ship registries	Flag state, ownership, management company, classification society, and change history
Corporate registries	Beneficial ownership of shell companies holding vessels, often in multiple jurisdictions
Commercial satellite imagery	Overhead views of transfer locations and anchorages
Port and customs publications	Arrival and departure records, cargo declarations where published
Insurance and industry notices	Withdrawal of cover, class suspensions, industry advisories
Trade press and sanctions notices	Designations, advisories, and enforcement actions

The investigative process.

1. **Baseline normal behavior.** Establish what ordinary voyages look like for a vessel class and route: continuous AIS transmission, plausible speeds, declared destinations matching arrivals.
2. **Detect anomalies.** Look for AIS gaps at sea (transponders switched off), positions that are physically implausible, identity broadcasts that conflict with registry data, and vessels reporting a location while imagery shows them elsewhere.
3. **Correlate gaps with meeting opportunities.** Where two vessels both go dark in overlapping

windows in the same general area, a ship-to-ship transfer becomes a hypothesis.

4. **Confirm with imagery.** Task or purchase satellite imagery over the suspected transfer area for the relevant window; two hulls alongside in open water is physical evidence that AIS silence cannot explain away.
5. **Follow the corporate chain.** Trace ownership and management through registry filings across jurisdictions, looking for recently formed entities, single-vessel companies, shared addresses, and repeated directors.
6. **Follow the service chain.** Identify which registries, class societies, and insurers appear repeatedly, since intermediaries are fewer in number than vessels and therefore better enforcement targets.
7. **Document and publish or refer.** Compile with full sourcing so findings can support designation, underwriting decisions, or press reporting.

Outcome. This body of work, produced by a mix of governments, commercial analytics firms, insurers, and journalists, supported vessel designations, insurance withdrawals, port entry refusals, and public advisories. The durable outcome is a change in the enforcement model: rather than pursuing individual cargoes, enforcement began targeting the small population of enabling intermediaries.

Transferable lesson. Anomalies in mandatory reporting systems are high-value findings. AIS exists for safety, not intelligence, and it is mandatory, which means silence is itself information. Any regulated system that requires disclosure produces the same opportunity: the exception is the signal. Look for what should be there and is not.

Discussion questions.

1. AIS silence is legal in some circumstances and is not proof of wrongdoing. How would you word an assessment that relies on it?
2. Which single source in the table above did the most work, and which was most necessary for confirmation? Are those the same source?
3. Why are intermediaries better targets than vessels, and what does that suggest about network analysis generally?

3.2 Law Enforcement: An Organized Retail Theft Network

Composite

The situation. A metropolitan area experienced a sharp rise in coordinated retail theft: crews entering stores, taking high-value goods in volume, and leaving quickly. Individual arrests were occurring, but the pattern continued. Detectives had 40-plus incident reports and no structure.

The requirement. How is stolen merchandise being aggregated and resold, and who is operating the aggregation layer? The decision it supported: whether to continue arresting crew members individually or to build a case against the resale operation.

What was collected.

Source	Material
Online marketplace listings	Bulk listings of the specific product categories being stolen, seller history, listing photographs
Listing photograph analysis	Backgrounds visible in seller photographs: shelving, flooring, fixtures, window views
Business records (public)	Business registrations, sales tax permits, fictitious business name filings
Property records	Ownership and tenancy of commercial units matching photograph backgrounds
Court records	Prior civil judgments, evictions, and case filings involving the same names and entities
Vehicle and traffic sources	Publicly reported vehicle descriptions across incident reports
Social media (public posts)	Public posts by crew members, including photographs showing merchandise and locations
Incident report analysis (internal)	Timing, method, target selection, and product category across all 40 reports

The investigative process.

1. **Pattern analysis first.** Before any online research, the incident reports were structured into a table: date, time of day, day of week, store, product category, crew size, method of entry and exit, vehicle description. Two clusters emerged, distinguished by product category and method.
2. **Product category as the search key.** The stolen categories were specific enough to search marketplaces for bulk resale, since ordinary consumers do not list forty units of one item.
3. **Photograph background analysis.** Several high-volume sellers used photographs taken in the same physical space. Distinctive fixtures and a partial view through a window allowed the space to be tentatively located.
4. **Records confirmation.** Property and business records for the candidate location identified a tenant business and its registered principal.
5. **Relationship mapping.** Court records and business filings connected that principal to two other entities, one of which had a shipping address that appeared in a separate incident report's vehicle registration.
6. **Corroboration before action.** Everything to this point was open source, uncorroborated by direct evidence, and therefore treated strictly as investigative lead material. Confirmation was pursued through lawful investigative process: surveillance, controlled purchase, and subpoenaed records.

7. **Case sequencing.** The prosecution strategy targeted the aggregation layer and the financial flow, with crew-level cases used to support the larger case rather than being disposed of individually.

Outcome. A single aggregation operation was identified as the outlet for both clusters. Charges targeted the operators and the business entities rather than only the crews, and incident volume in the affected retail district declined substantially over the following two quarters.

Transferable lesson. OSINT produces leads, not evidence. Every finding above was used to decide where to point lawful investigative process. Nothing was presented as proof. This distinction is not a technicality: intelligence used as evidence without independent confirmation is a due process problem and a case-losing error. Chapter 1's separation of intelligence from evidence applies literally here.

Discussion questions.

1. Photograph background analysis located a physical space. What error would a careless analyst make at that step, and how would corroboration catch it?
2. Why was pattern analysis of the existing reports performed before any online searching? What would have gone wrong in the reverse order?
3. What civil liberties safeguards apply to the collection of public social media posts here, and how would you document compliance?

3.3 Cybersecurity: External Exposure and a Brand Impersonation Campaign

Composite

The situation. A regional financial services firm with 900 employees had a mature internal security program and no visibility outside its own perimeter. Customers had begun reporting phishing messages that appeared to come from the firm.

The requirement. Two: what internet-facing assets and exposures does an attacker see when they look at us, and what infrastructure is currently being used to impersonate our brand? The decisions supported: remediation prioritization and takedown action.

What was collected.

Source	Material
Certificate transparency logs	Every publicly trusted TLS certificate issued for names resembling the firm's domains
Domain registration data	Registration dates, registrars, and name server patterns for lookalike domains
Passive DNS	Historical resolution data linking domains to hosting infrastructure

Source	Material
Internet-wide scan data	Publicly reachable services, versions, and exposed administrative interfaces on the firm's own address space
Public code repositories	Accidentally committed configuration files, API keys, and internal hostnames
Credential leak notification services	Corporate email addresses appearing in third-party breach corpora
Job postings (the firm's own)	Technology stack details published in requirements, revealing internal systems
Social and professional profiles	Employee-published detail about internal tooling and reporting structure
App stores and marketplaces	Unauthorized applications using the firm's name and logo

The investigative process.

1. **Enumerate the legitimate footprint first.** Establish what the firm actually owns: domains, address ranges, cloud tenancy, subdomains. This step is skipped constantly, and skipping it makes the rest of the exercise unreliable, because you cannot recognize an impostor without knowing the genuine set.
2. **Certificate transparency sweep.** Query logs for names containing the brand and common visual substitutions. Certificate issuance is often the earliest public signal of a phishing site, frequently appearing before the site is operational.
3. **Registration and infrastructure clustering.** Group the lookalike domains by registrar, registration date, name server, and hosting. Clusters indicate one operator rather than many.
4. **Historical resolution analysis.** Passive DNS revealed that four domains had at different times resolved to the same host, connecting a campaign that appeared to be four separate incidents.
5. **Own-exposure assessment.** Scan data and repository searches identified an exposed administrative interface on a forgotten test system and a committed credential in a public repository. Both were remediated the same week.
6. **Human exposure assessment.** Job postings and professional profiles were reviewed for detail that assisted attackers, and guidance was issued to reduce specificity without harming recruiting.
7. **Action.** Takedown requests filed with registrars and hosts, detection rules deployed for the identified infrastructure, customer advisory issued, and a standing monitoring requirement established for new certificate issuance matching the brand.

Outcome. Nineteen impersonating domains identified, of which fourteen were removed within three weeks. Two genuine exposures on the firm's own infrastructure were closed. The campaign was

assessed as a single operator based on infrastructure clustering, which changed the response from four small incidents to one campaign with a coherent countermeasure. Continuous certificate monitoring subsequently identified new domains before they were used.

Transferable lesson. Public infrastructure records were not designed as intelligence sources, and they are among the best available. Certificate transparency exists to make certificate issuance auditable, not to catch phishing, and it catches phishing very effectively. Whenever a system is designed for transparency or accountability, it produces analyzable open data. Learn what those systems are in your domain.

Discussion questions.

1. Why is enumerating the legitimate footprint the mandatory first step? Give a concrete example of an error caused by skipping it.
 2. The firm's own job postings assisted attackers. How would you balance recruiting effectiveness against exposure reduction?
 3. Infrastructure clustering supported an assessment that one operator was responsible. What confidence level would you assign and why? What would raise it?
-

3.4 Journalism: Locating Undisclosed Detention Facilities From Satellite Imagery

Historical

The situation. Reporting from the late 2010s alleged large-scale construction of internment facilities in the Xinjiang region of China. The government denied the characterization, and independent access for journalists was severely restricted. Establishing the scale of construction from the ground was not possible.

The requirement. How many such facilities exist, where are they, and what is their scale and construction timeline? The consumers were the public, policymakers, and human rights bodies.

What was collected and the investigative process.

1. **An anomaly in a mapping service.** Investigators noticed that a widely used Chinese mapping platform displayed blank or masked tiles over certain locations, while the same coordinates on other providers showed imagery. The masking was intended to conceal, and it inadvertently produced a catalog of coordinates of interest.
2. **Systematic extraction of masked coordinates.** The blank tiles were enumerated across the region, generating tens of thousands of candidate locations.
3. **Filtering.** Most masked tiles corresponded to ordinary sensitive infrastructure. Candidates were filtered against other imagery providers and reduced to a manageable set.
4. **Visual classification.** Remaining sites were examined in higher-resolution commercial imagery for the physical signatures of a detention facility: perimeter walls with guard towers, internal

segmentation, restricted single-entry access, high-density barracks-style buildings, and industrial buildings adjacent to secured compounds.

5. **Temporal analysis.** Archived imagery from successive years established construction start dates and expansion sequences, converting a static count into a timeline.
6. **Corroboration.** Findings were cross-referenced against government procurement and construction tender documents published online, official budget line items, satellite-visible expansion matching tender dates, and testimony from former detainees who were able to describe layouts.
7. **Publication with full coordinates.** The investigation published locations and imagery so that any reader or rival newsroom could verify each site independently.

Outcome. A published database of hundreds of identified facilities with coordinates, imagery, and construction timelines. The work was widely cited by governments, parliaments, and international bodies, and it substantially shifted the public argument from whether facilities existed to what they were. Because the underlying evidence was published, denials had to contend with imagery any reader could check.

Transferable lesson. Two, and both are important.

Concealment leaves a signature. The act of hiding produced the very index used to find. This generalizes: redactions with visible lengths, blurred regions, removed pages with numbered gaps, and access denials all mark the location of something. Look for the shape of the absence.

Publishing the evidence is a verification strategy. Releasing coordinates invited adversarial checking. Surviving that checking is what made the work authoritative. Compare this to the failure cases in Section 1.6, where claims were not traceable.

Discussion questions.

1. The masked-tile technique depended on a specific implementation detail that could be changed at any time. What does that imply about tool-dependence versus method-dependence in OSINT?
2. Visual classification required distinguishing detention facilities from prisons, factories, and schools with walls. How would you write down the classification criteria so a second analyst could apply them consistently?
3. What role did non-open sources (former detainee testimony) play, and how does that fit the all-source model from Section 1.5.2?

3.5 Disaster Response: Crisis Mapping After a Major Earthquake

Historical

The situation. A catastrophic earthquake struck Haiti in January 2010, destroying much of the capital's infrastructure. Responders arrived to find that existing maps of Port-au-Prince were sparse, out of date,

or unavailable, and that the road network, building locations, and camp sites could not be reliably referenced.

The requirement. Where are the affected populations, which routes are passable, where are the medical and shelter needs, and what does the damaged city actually look like right now? Consumers: search and rescue teams, medical organizations, logistics planners, and the affected population.

What was collected.

Source	Material
Newly released satellite and aerial imagery	Post-event high-resolution imagery released for humanitarian use by commercial and government providers
Volunteer digitization	Thousands of remote volunteers tracing roads, buildings, and features into an open mapping platform
SMS and social reporting	Text messages and social posts from the affected population, submitted to a public reporting platform via a short code
Translation volunteers	Diaspora volunteers translating Haitian Creole messages in near real time
Radio and local media	Local broadcast information on conditions and needs

The investigative process.

1. **Imagery released and imported.** Providers made post-event imagery available, which was loaded into an open mapping platform.
2. **Distributed tracing.** Volunteers worldwide traced features from imagery. Within weeks, the map of the affected area went from sparse to among the better-mapped areas in the region.
3. **Structured public reporting.** A short code was publicized locally, allowing residents to text needs and locations. Messages were received, translated, geolocated where possible, categorized (trapped persons, medical, water, food, security), and plotted.
4. **Verification and triage.** Reports were assessed for specificity and corroboration. Duplicate and unlocatable reports were flagged rather than discarded, and categories were prioritized.
5. **Delivery to responders.** Map data and report feeds were made available in formats usable by field teams, including on devices with intermittent connectivity.

Outcome. Responding organizations, including military and civilian search teams, used the volunteer-produced map data operationally. The effort is widely credited with directing rescues and aid deliveries and with demonstrating that distributed volunteer effort could produce operationally usable geographic intelligence at speed. It became the founding example of modern crisis mapping, and the model was repeated in later disasters, including large United States flooding events where volunteer mapping and

civilian rescue coordination operated alongside official response.

The honest complications. Reviews of this and later efforts identified real problems that students should carry forward:

- **Volume without triage overwhelms responders.** Thousands of unverified reports can consume field capacity rather than direct it.
- **Coordination with official structures was informal,** which limited how far the data could be trusted for life-safety decisions.
- **Reporting is biased toward those able to report,** which under-represents exactly the most vulnerable populations, an instance of the availability bias in Section 1.4.2.
- **Expectation management matters.** Publicizing a reporting channel creates an expectation of response that volunteers cannot guarantee.

Transferable lesson. In the first hours of a disaster, open sources are frequently the only sources, and the constraint is triage rather than collection. The professional contribution is not gathering more reports. It is categorizing, geolocating, deduplicating, and prioritizing them so that scarce responders are pointed correctly. That is processing and analysis, exactly as Chapter 1 describes.

Discussion questions.

1. This effort produced enormous volume rapidly. Design a triage rule set that a volunteer with 30 seconds per report could apply.
2. Reporting bias under-represents the most vulnerable. How would you word a situational awareness product to keep that limitation in front of the consumer?
3. Volunteer-produced data was used for life-safety decisions. What verification standard would you require before allowing that, and who should set it?

3.6 Business Investigations: Pre-Acquisition Due Diligence

Composite

The situation. A mid-sized manufacturer planned to acquire a smaller competitor for \$34M. The seller's disclosures were clean. The buyer's counsel commissioned open source diligence with four weeks before signing.

The requirement. Does the public record contradict or complicate the seller's representations regarding litigation, ownership, environmental compliance, key-person dependency, and customer concentration? The decision: proceed, renegotiate, or walk.

What was collected and found.

Source	Finding
Court dockets, multiple counties and federal	Two employment matters disclosed; one product

Source	Finding
	liability suit filed 14 months earlier under a subsidiary name not listed in the disclosure schedule
Corporate registries across three states	The subsidiary in question was under common ownership but omitted from the corporate structure chart provided
UCC filings	A lien on production equipment held by a lender not disclosed in the debt schedule
Environmental agency records	An open notice of violation at the primary plant, with a compliance deadline falling three months after the planned closing
Commercial satellite imagery	Outdoor storage at the plant expanding across two years, consistent with the practice described in the violation notice
Building permits	No permit filed for a facility improvement the seller described as substantially complete
Professional profiles and job postings	The named lead engineer had updated their profile status; simultaneously, the seller was quietly recruiting for that role
Customer public disclosures	A customer representing an estimated significant share of revenue had publicly announced a supplier consolidation program
Trade press and local news	A prior local dispute over plant emissions, with community opposition on record
Import and shipping records	Supplier concentration inconsistent with the diversified supply base described

The investigative process.

1. **Start from the representations.** Diligence OSINT is unusual in that the requirement is handed to you: the seller has made specific claims, and your job is to test each one against the public record. Each representation became a line item.
2. **Entity expansion before searching.** Every name variant, former name, subsidiary, and registered agent address was enumerated first. The critical finding depended on this step, because the undisclosed litigation was filed against a name not on the provided list.
3. **Jurisdiction mapping.** Records searched in every jurisdiction where the company operated, incorporated, or held property, rather than only the headquarters state.
4. **Physical verification.** Imagery and permit records were used to test claims about facilities, since a claim about a completed improvement is checkable.

5. **Key-person exposure.** Public professional signals were read together with the company's own recruiting activity, which is a stronger indicator in combination than either alone.
6. **Documentation for negotiation.** Every finding was documented to the standard in Section 1.11.5, because findings used in a negotiation will be challenged by the other side's counsel.

Outcome. The buyer proceeded at a reduced price with a larger escrow, specific indemnities for the undisclosed litigation and the environmental matter, and a retention package for the engineer. The findings did not kill the deal; they repriced it. The environmental notice alone represented a remediation cost approaching a meaningful fraction of the discount obtained.

Transferable lesson. Entity expansion is the highest-leverage step in corporate OSINT, and it is the one most often skipped. Searching the name you were given finds what you were meant to find. The material findings live under former names, subsidiaries, registered agent addresses, and jurisdictions nobody mentioned.

Discussion questions.

1. The critical finding depended on searching a name the seller did not provide. Was the omission necessarily deliberate? How would you word the finding without alleging intent?
2. Which findings are facts, and which are assessments? Rewrite three of the table rows to make the distinction explicit.
3. Imagery corroborated a records finding. Why is that pairing stronger than either source alone?

3.7 Consumer Protection: Recognizing a Fraudulent Online Retailer

Composite, using patterns documented by consumer protection agencies

The situation. A consumer sees a heavily advertised online store offering a sought-after product at 60% below normal retail, with a countdown timer and hundreds of positive reviews.

The requirement. Is this a legitimate retailer? The decision is immediate and personal: enter payment details or not. This case is included because it is the OSINT application every student will use within a month of finishing this chapter.

The investigative process, in the order a consumer can actually perform it.

Step	Method	Signal in this case
1	Check domain registration date	Registered 19 days ago; legitimate retailers with this inventory rarely have three-week-old domains
2	Look for a physical address and phone	Address given is a residential unit in an unrelated country; phone not answered

Step	Method	Signal in this case
3	Reverse image search the product photographs	Identical images appear on the manufacturer's site and on three other stores with different names
4	Reverse image search the "team" photographs	Stock photography, appearing on unrelated sites
5	Read the policy pages closely	Returns policy contradicts itself on the window; text appears copied, with another company's name left in one paragraph
6	Examine the reviews	Hundreds posted within a few days, similar sentence structure, no mid-range ratings, no reviews mentioning specific product details
7	Search the store name plus complaint terms	Complaints on consumer forums within the past two weeks describing charges without delivery
8	Check regulator and consumer agency resources	Store name appears in a recent agency advisory pattern description, though not by name
9	Check the payment methods offered	No conventional card processing with dispute rights; pressure toward irreversible transfer methods
10	Check certificate and infrastructure	Certificate issued the same week as registration; hosting shared with several similarly named stores registered in the same window

The finding. Assessed with high confidence as a fraudulent storefront, based on ten independent indicators, of which any three would be sufficient. The strongest single indicator is item 10: infrastructure clustering with multiple similarly recent, similarly named stores, because a single operator running many disposable storefronts is the actual business model.

Outcome. No purchase. Reported to the relevant consumer protection agency and to the hosting provider. In the composite scenario, the site was gone within a month, which is itself the pattern: these operations are designed to be short-lived, which is why domain age is such a strong signal.

The reverse application: protecting others. The same method identifies impersonation of your own employer, fraudulent rental listings (search the listing photographs; they are usually copied from a genuine listing elsewhere), fake charity appeals after disasters, and employment scams.

Transferable lesson. Convergence of weak indicators is a strong finding. No single item in that table proves fraud. Domain age alone proves nothing; every legitimate new business has a new domain. Ten independent indicators pointing the same direction, from sources that a fraudster cannot cheaply coordinate, produce high confidence. This is exactly the reasoning pattern from Section 1.7.5, applied to a \$200 decision instead of an international one, and the logic does not change with the stakes.

Discussion questions.

1. Which three of the ten indicators would you keep if you had 90 seconds? Defend the selection on the basis of diagnostic power, not ease.
2. A legitimate new small business would fail several of these tests. How do you avoid harming legitimate sellers with this method?
3. Write the assessment as a two-sentence product with a confidence level, addressed to a family member who does not know what OSINT is.

3.8 Cross-Case Comparison

Case	Domain	Key sources	Decisive technique	Outcome type
Shadow fleet	National security	AIS, registries, imagery	Anomaly detection in mandatory reporting	Enforcement targeting shifted to intermediaries
Retail theft network	Law enforcement	Marketplace listings, property and court records	Photograph background geolocation, then lawful process	Case built against the aggregation layer
Brand impersonation	Cybersecurity	Certificate transparency, passive DNS, scan data	Infrastructure clustering	Nineteen domains identified; own exposures closed
Detention facilities	Journalism	Masked map tiles, commercial imagery, tenders	Using the signature of concealment	Public database; argument reframed
Earthquake mapping	Disaster response	Released imagery, volunteer tracing, public SMS	Distributed processing and triage	Operationally used map and needs data
Acquisition diligence	Business investigation	Dockets, registries, UCC, imagery, permits	Entity expansion before searching	Deal repriced with indemnities
Fraudulent storefront	Consumer protection	Registration data, reverse image search, reviews,	Convergence of independent weak indicators	Loss avoided; report filed

Case	Domain	Key sources	Decisive technique	Outcome type
		hosting		

Two patterns run through the whole table. First, **the decisive technique is almost never a tool; it is a way of thinking about where information is structurally likely to leak.** Anomalies in mandatory systems, signatures of concealment, clustering of shared infrastructure, and name variants nobody mentioned. Second, **every case pairs an open source finding with an independent check:** imagery against records, records against physics, one publisher against another. The pairing is what makes the finding survive contact with a skeptic.

Part 4: Chapter Summary

OSINT is intelligence, not searching. Open source intelligence is intelligence produced from publicly available, lawfully obtainable information collected without defeating any protection the owner placed on it. The word "intelligence" carries the full Chapter 1 weight: a requirement, source evaluation, analysis, an explicit judgment, a stated confidence level, and timely delivery to a decision maker. A compilation of public material is open source information, and calling it intelligence is the discipline's most common professional error.

"Open" is defined by authorization, not convenience. Publicly available does not mean free, easy, or online. It means obtainable by any member of the public through lawful means without misrepresenting identity or circumventing access controls. Four tests apply before collection: access, identity, protection, and purpose and harm. The fourth test is the one that requires judgment, because aggregating individually public facts about a private person can produce something that person never made public. That is the mosaic effect, and it is the discipline's central ethical problem.

The history shows a recurring cycle. Systematic use of public material runs from pre-modern commerce through wartime broadcast monitoring, Cold War institutionalization, commercial digitization, the web, social media and mobile, the verification era of independent open source investigation, and the current period of mass OSINT and generative content. Four lessons recur: the value was recognized long before the investment followed; every capability expansion brought a proportional deception risk; the bottleneck moved from access to verification; and capability decentralized from states to small teams and individuals.

The evolution was driven by six specific changes. Publication cost collapsed, sensors became consumer goods, commercial satellite imagery matured, machine-readable public infrastructure appeared, verification methods were formalized into teachable techniques, and the practitioner community became public and adversarially peer-reviewed. What did not change: the requirement still comes first, source evaluation is still two separate questions, and a person still has to own the judgment.

OSINT matters for seven structural reasons and is limited by six. Advantages: cost, speed, shareability, accessibility, auditability, tipping and cueing, and breadth. Limitations: volume,

deception, circular reporting, availability bias, weak reach into intent, and ephemerality. Shareability and auditability deserve particular attention, because they are unique among the disciplines: an OSINT finding can be shown to a jury, a journalist, a partner, or the public, and it can be reproduced by a skeptic.

Modern practice is open-source-first. Exhaust what is publicly knowable before spending scarce collection capacity, both to answer the requirement and to narrow what remains into a precise tasking for expensive collection. OSINT frames the problem; other disciplines reach what the frame cannot. The three current challenges are synthetic content (including the liar's dividend, where authentic evidence can be dismissed as fake), commercially available data on individuals, and platform access volatility, which is why the durable skills are method-level rather than tool-level.

Failures are usually not collection failures. Failure modes divide into collection, processing, analysis, dissemination, and consumer failures, and the remedies are entirely different. Across Pearl Harbor, the 2002 Iraq assessment, the 2013 crowdsourced misidentification, and the ordinary corporate pattern of an unwritten requirement, the information was substantially available in every case. The distinguishing failures were fragmentation, assumption treated as fact, layering, absent alternatives, miscalibrated confidence, and dissemination that could not be recalled.

Successes share a recognizable profile. Anchoring in physical or documentary fact, genuinely independent sources, total sourcing transparency, specificity over generality, robustness under adversarial scrutiny, and ethical grounding in public-interest subject matter. The successes did not have better access than the failures. They had verification thresholds and accountable analysts.

The three user communities face different constraints, and the counterintuitive one is government. Governments have the deepest resources, the ability to fuse with non-open sources, and the legal authority to act on findings, and they also face the strictest limits on collecting about individuals: constitutional protections, criminal predicate, retention limits, audit, and oversight. Commercial users face privacy law, computer misuse law, terms of service, trade secret law, consumer reporting rules, and sector regulation, and their characteristic failure is that the requirement was never written. Personal users have almost no formal constraint, which makes personal use the setting where individual ethics carry the entire weight.

Verification is the tradecraft. Establish provenance before assessing truth: who published first, when, on what basis, whether the content has appeared before, and what interest the publisher has. The technique set includes reverse image search, geolocation from fixed features, chronolocation from shadows and weather, metadata examination, cross-platform correlation, account history analysis, satellite comparison, documentary corroboration, and physical consistency checks. Against synthetic content, deprioritize artifact hunting and prioritize provenance and corroboration, because an event that actually happened generates an evidentiary context that a fabrication does not.

Documentation is not optional. Every finding carries a claim, a source, a URL, an access date and time, an archived copy, an independence assessment, reliability and credibility ratings, the verification steps performed, and the analyst's initials. Without that trail, a finding is an opinion. With it, a finding is something a skeptic can test, which is the entire basis of OSINT's credibility.

And the ethics are load-bearing. No private individuals. No circumvention. No false identities. No profiles built for curiosity. Convergence of weak indicators is a strong finding, and irreversible publication of an unverified identification is a permanent harm. The capability taught in this chapter is cheap, powerful, and unsupervised, and the person deciding whether to use it responsibly is you.

Part 5: Key Terms

Grouped by theme, with an alphabetical index at the end. Terms carried forward from Chapter 1 are marked (Ch.1) and restated briefly.

5.1 Core Definitions

Open source intelligence (OSINT). Intelligence produced from publicly available, lawfully obtainable information collected without defeating any protection the owner placed on it.

Open source data (OSD). Raw, unprocessed public material in its original form.

Open source information (OSIF). Public material that has been collected and given basic context but not evaluated or analyzed against a requirement.

Validated OSINT (OSINT-V). OSINT corroborated against non-open sources or established with very high certainty.

Publicly available information (PAI). Information any member of the public can lawfully obtain. Cost and difficulty do not disqualify it; circumvention does.

Commercially available information (CAI). Data lawfully purchasable from commercial vendors, including aggregated personal data. Legally distinct from PAI and increasingly subject to policy and statutory restriction.

Publicly available does not mean authorized. The principle that the public nature of a source answers a question about the source, not about the collector's authority to collect, retain, or use it.

Intelligence (Ch.1). Information collected against a requirement, evaluated, analyzed in context, given a judgment and a confidence level, and delivered to a decision maker in time to matter.

All-source analysis (Ch.1). Analysis fusing multiple collection disciplines so each covers the others' weaknesses.

Open-source-first. The practice of exhausting publicly knowable information before tasking expensive collection, both to answer the requirement and to narrow what remains.

Tippling and cueing. Using one source to identify where to point another, more expensive one.

5.2 Sources and Collection

Gray literature. Material that is publicly available but not formally published or indexed: conference slides, marketing collateral, technical manuals, internal-looking documents posted openly.

AIS (Automatic Identification System). A mandatory maritime broadcast system transmitting vessel identity, position, course, and speed. Publicly receivable and aggregated by commercial and free platforms.

ADS-B (Automatic Dependent Surveillance-Broadcast). An aviation equivalent broadcasting aircraft identity and position.

AIS gap / going dark. A period during which a vessel stops transmitting. Sometimes lawful and innocuous, sometimes an indicator of concealment.

Ship-to-ship transfer. Transferring cargo between vessels at sea, which can obscure cargo origin.

Certificate transparency log. A public, append-only record of publicly trusted TLS certificates issued. Frequently the earliest public signal that a lookalike domain has been created.

Passive DNS. Historical records of domain name resolution, used to link domains to infrastructure over time.

Internet-wide scan data. Published results of systematic scanning of internet-reachable services, showing exposed systems and versions.

Corporate registry. A government record of company formation, officers, agents, and status.

Beneficial ownership. The natural persons who ultimately own or control an entity, as distinct from the entity named on a filing.

UCC filing. A public record of a security interest in personal property, useful for identifying undisclosed lenders and encumbered assets.

Docket. The public record of filings and proceedings in a court case.

Entity expansion. Enumerating all name variants, former names, subsidiaries, registered agents, and jurisdictions associated with a subject before beginning record searches. The highest-leverage step in corporate OSINT.

SOCMINT (Ch.1). Social media intelligence; collection from social platforms, a sub-practice of OSINT with distinct privacy considerations.

Research account. An account used for collection that does not misrepresent the researcher's identity. Distinct from a false persona, which is outside OSINT.

5.3 Verification Tradecraft

Provenance. The origin and chain of custody of a piece of content: who published it first, when, and on what basis.

Geolocation. Determining where an image or video was captured, typically by matching fixed physical features against street-level or satellite imagery.

Chronolocation. Determining when content was captured, typically from shadow direction and length against solar position, weather records, or vegetation state.

Reverse image search. Querying an image against indexed collections to find earlier or other appearances, primarily to detect recycled content.

Recycled content. Authentic material from an earlier event republished as though it depicts a current one. The most common form of open source deception.

Change detection (Ch.1). Comparing imagery of the same location across time to identify what changed and when.

Metadata. Data embedded in a file describing its creation: device, timestamp, and sometimes coordinates. Corroborating rather than conclusive, since it can be stripped or edited.

Corroboration (Ch.1). Independent confirmation by a source that did not derive from the first.

Circular reporting (Ch.1). Apparent corroboration created when multiple sources all trace to a single unverified origin.

Source reliability (Ch.1). Assessment of a provider's track record, rated A through F.

Information credibility (Ch.1). Assessment of a specific claim's plausibility on its own merits, rated 1 through 6.

Coordinated inauthentic behavior. Activity by accounts or outlets that presents manufactured or centrally directed content as independent and organic.

Synthetic content. Text, image, audio, or video generated by a model rather than captured from reality.

Liar's dividend. The advantage gained by a bad actor when the existence of plausible fakes allows authentic evidence to be dismissed as fabricated.

Physical consistency check. Testing content against physics: sound delay versus distance, shadow geometry, reflections, scale, and weather consistency. The strongest available check where applicable.

Archiving at collection. Preserving a copy of source material at the moment it is collected, because online content is ephemeral.

Link rot. The progressive disappearance or alteration of previously cited online material.

Collection log. The documented trail of every finding, its source, access time, verification steps, and ratings.

5.4 Analysis and Assessment

Requirement (Ch.1). A specific question that must be answered to support a decision.

Indicator (Ch.1). An observable whose presence or absence helps answer a requirement.

Confidence level (Ch.1). A stated judgment about how solid the basis for an assessment is. Distinct from probability.

Estimative language (Ch.1). Standardized probability vocabulary used so readers interpret uncertainty consistently.

Intelligence gap (Ch.1). A specific thing not known, stated explicitly, with its effect on the assessment.

Convergence of weak indicators. The principle that multiple independent low-strength indicators pointing the same direction can support high confidence, provided they could not be cheaply coordinated by an adversary.

Availability bias (OSINT context). The systematic over-representation in open sources of connected populations, permissive environments, dominant languages, urban areas, and photogenic events. Makes absence of reporting weak evidence of absence.

Mosaic effect. The creation of a sensitive whole from individually non-sensitive public parts, particularly when aggregating facts about an individual.

Layering (Ch.1). Treating an earlier product's assumption or low-confidence judgment as established fact in a later product.

Confirmation bias (Ch.1). Noticing and crediting evidence that supports an existing belief while discounting contrary evidence.

Failure mode taxonomy. The classification of intelligence failure as collection, processing, analytic, dissemination, or consumer failure, each with a different remedy.

Leads versus evidence. The principle that OSINT findings direct lawful investigation and do not themselves prove facts, particularly in legal contexts.

5.5 Legal and Ethical Terms

Terms of service (ToS). Contractual conditions imposed by a platform, which may restrict automated collection, account creation, or data use.

Computer misuse law. Statutes prohibiting unauthorized access to computer systems, such as the Computer Fraud and Abuse Act in the United States.

GDPR. The European Union General Data Protection Regulation, governing processing of personal data, including some processing of publicly posted personal data.

Lawful basis. Under data protection regimes, the legal ground required to process personal data.

Purpose limitation. The principle that data collected for one stated purpose may not be freely repurposed.

Retention limit. A defined period after which records must be purged.

Criminal predicate (Ch.1). A reasonable suspicion of criminal involvement, required in many jurisdictions before an individual may be the subject of retained intelligence records.

Fair Credit Reporting Act (FCRA). United States statute regulating information used in decisions about employment, credit, insurance, and housing, with notice, accuracy, and dispute obligations.

Trade secret (Ch.1). Information with commercial value that is not generally known and that its owner took reasonable measures to protect.

Doxxing. Publishing identifying information about an individual, typically to expose them to harassment or harm. Public availability of each component is not a defense.

Pretexting. Obtaining information by misrepresenting identity or purpose. Outside OSINT by definition.

False persona. A fabricated identity used to obtain access. Distinct from a research account and outside OSINT.

Researcher OPSEC. Managing the researcher's own detectable footprint during collection, since some collection is visible to the subject.

Prohibited-activity list. A written enumeration of collection activities an organization will not perform, decided in advance rather than under deadline pressure.

5.6 Alphabetical Index

ADS-B · AIS · AIS gap · All-source analysis · Archiving at collection · Availability bias · Beneficial ownership · CAI · Certificate transparency log · Change detection · Chronolocation · Circular reporting · Collection log · Commercially available information · Computer misuse law · Confidence level · Confirmation bias · Convergence of weak indicators · Coordinated inauthentic behavior · Corporate registry · Corroboration · Criminal predicate · Docket · Doxxing · Entity expansion · Estimative language · Failure mode taxonomy · Fair Credit Reporting Act · False persona · GDPR · Geolocation · Gray literature · Indicator · Information credibility · Intelligence · Intelligence gap · Internet-wide scan data · Layering · Lawful basis · Leads versus evidence · Liar's dividend · Link rot · Metadata · Mosaic effect · Open-source-first · Open source data · Open source information · Open source intelligence · OSINT-V · Passive DNS · Physical consistency check · Pretexting · Prohibited-activity list · Provenance · Publicly available information · Purpose limitation · Recycled content · Requirement · Research account · Researcher OPSEC · Retention limit · Reverse image search · Ship-to-ship transfer · SOCMINT · Source reliability · Synthetic content · Terms of service · Tipping and cueing · Trade secret · UCC filing · Validated OSINT

Part 6: Knowledge Assessment

6.1 Short Answer (20 questions)

Answer in three to five sentences unless otherwise noted.

1. Define OSINT and explain which of the three words in the term does the most work in distinguishing it from ordinary research.
2. Explain the difference between open source data, open source information, and open source intelligence, with an example of each.
3. State the four tests for whether a source is "open" and explain which test requires judgment rather than a legal determination.

4. A document is accessible on a misconfigured public server. Is it an open source? Explain your reasoning.
5. Describe three eras in the history of OSINT and identify what changed the discipline in each.
6. Explain the status paradox that persisted through much of the Cold War period, and state whether you think it has been resolved.
7. Identify the six changes that drove OSINT from a supporting activity to a primary discipline, and state which one you consider most consequential.
8. Explain the claim that the bottleneck in OSINT moved from access to verification. Give a concrete comparison across two eras.
9. List four structural advantages of OSINT and explain why shareability is more valuable than students typically assume.
10. List four structural limitations of OSINT and explain why absence of open reporting is weak evidence of absence.
11. Explain what open-source-first means and why the narrowing step is the source of its value.
12. Describe how OSINT contributes to two other collection disciplines, with a specific example for each.
13. Explain the liar's dividend and why it makes verification harder than simply detecting fakes.
14. Name the five intelligence failure modes and state a different remedy for each.
15. Using the 2002 Iraq WMD case, name four specific analytic failures and state the structured technique that addresses each.
16. Using the 2013 crowdsourced misidentification case, explain what professional controls were missing and why publication made the failure irreversible.
17. Identify four features common to the OSINT success cases in this chapter, and explain why access was not one of them.
18. Explain why governments face stricter constraints than companies on collecting about individuals, despite having greater resources and authority.
19. Explain the mosaic effect and why "each fact was public" is not an adequate ethical defense.
20. State the documentation standard for an OSINT finding and explain what is lost when it is not followed.

6.2 Multiple Choice (15 questions)

1. Which statement best describes what makes a source "open"? a) It is free of charge b) It is available on the internet c) It can be lawfully obtained by any member of the public without circumventing protections d) It is indexed by a search engine
2. A 40-page compilation of everything publicly known about a company, with no judgment or

confidence level, is best described as: a) OSINT b) Open source information c) Validated OSINT d) All-source intelligence

3. Creating a fake profile to view a restricted account is: a) Standard OSINT tradecraft b) Acceptable if the purpose is legitimate c) Outside OSINT, because it requires misrepresenting identity d) Acceptable if the platform's terms permit research

4. During World War II, analysts derived industrial and logistics estimates from which kind of material? a) Intercepted encrypted communications only b) Publicly available broadcasts, publications, statistics, and timetables c) Aerial photography exclusively d) Recruited human sources exclusively

5. The most common form of deception in open source material is: a) Entirely fabricated documents b) Authentic content from an earlier event republished as current c) Manipulated satellite imagery d) Falsified corporate filings

6. Which of the following is the strongest verification technique when it can be applied? a) Looking for visual artifacts suggesting generation b) Checking whether the content obeys physics, such as sound delay against distance c) Counting how many outlets carried the claim d) Checking whether the account has many followers

7. Certificate transparency logs are valuable for OSINT because: a) They record private keys b) They provide a public, append-only record of certificate issuance, often revealing lookalike domains before use c) They list all domains ever registered d) They identify the operator of any domain

8. In the maritime sanctions case, why is a gap in AIS transmission analytically useful? a) It proves illegal activity b) Transmission is expected, so its absence is an anomaly worth investigating c) It reveals the cargo manifest d) It is always unlawful

9. Across the failure cases in this chapter, the information needed was: a) Almost never available b) Available only through classified channels c) Substantially available in nearly every case d) Available but legally unobtainable

10. Which failure mode describes a correct assessment that never reached the decision maker in usable form? a) Collection failure b) Analytic failure c) Dissemination failure d) Processing failure

11. Entity expansion in corporate OSINT means: a) Growing the size of the investigation team b) Enumerating name variants, former names, subsidiaries, agents, and jurisdictions before searching c) Expanding the geographic scope of imagery d) Adding more search engines

12. Which of these is the most appropriate personal use of OSINT? a) Researching a former partner's current address b) Compiling a profile of a neighbor you dislike c) Verifying that a contractor is licensed and has no pattern of judgments before hiring them d) Investigating a coworker on a friend's behalf

13. The mosaic effect refers to: a) Combining imagery from multiple satellites b) Assembling individually public facts into a sensitive whole the subject never made public c) Using multiple search engines for the same query d) A pattern of tiles used in geolocation

14. In the Xinjiang detention facility investigation, the initial index of locations came from: a) A leaked

government document b) Blank or masked tiles on a mapping platform, which marked the locations being concealed c) Recruited human sources d) Intercepted communications

15. Convergence of weak indicators supports high confidence when: a) There are simply many indicators b) The indicators come from sources an adversary could not cheaply coordinate c) The indicators all appear on social media d) A trusted outlet reported all of them

6.3 True or False (10 questions)

Mark true or false. For each false statement, write one sentence explaining what makes it false.

1. If information is accessible on the internet, it is lawful and ethical to collect and use it.
 2. OSINT means free information, so any collection with a cost is by definition not OSINT.
 3. Open source findings can be shared and independently reproduced in ways that classified findings generally cannot.
 4. Governments generally face fewer restrictions than private companies on collecting information about individuals from public sources.
 5. In most documented intelligence failures, the primary problem was that the necessary information had not been collected.
 6. Metadata embedded in a photograph is conclusive proof of when and where it was taken.
 7. Absence of open source reporting about an event is strong evidence that the event did not occur.
 8. Publishing an identification of a private individual based on uncorroborated open source material is a recoverable error, since corrections can be issued.
 9. OSINT findings used in a criminal investigation should be treated as investigative leads requiring independent confirmation rather than as evidence.
 10. Under some data protection regimes, collecting and storing publicly posted personal data can be a regulated processing activity requiring a lawful basis.
-

Part 7: Critical Thinking Exercises

For each scenario, answer the questions posed. Where the exercise asks you to identify a failure, name the failure mode from Section 1.6.1. Where it asks about appropriateness, apply the four tests from Section 1.1.4.

7.1 Is OSINT appropriate? A hiring manager asks you to "look into" three finalists for a role, including their social media. Assess the request. What is legally risky, what is ethically problematic, and what would a defensible version of this activity look like?

7.2 Is OSINT appropriate? A regional water utility asks you to determine whether foreign state actors have the capability and intent to disrupt its control systems. Which part of that requirement can OSINT

answer, which part cannot, and how would you rewrite the requirement so that it is answerable?

7.3 Source reliability. Four accounts post the same claim about an industrial accident within 20 minutes. Two are local residents with years of unrelated posting history. Two are accounts created in the last month that post only about this industry. How do you rate reliability and credibility, and what does the timing pattern tell you?

7.4 Source reliability. A finding rests on a document posted to a file-sharing site by an anonymous account, which appears internally consistent and matches other reporting. Write the sentence you would use in a product to characterize this source, and state what you would need before raising confidence.

7.5 Circular reporting. Eleven outlets report a specific casualty figure. Nine cite "officials," one cites a named agency, and one cites another outlet. Describe your next three research steps and what each would resolve.

7.6 Recognizing failure. A company's threat intelligence team produced a correct assessment three weeks before an incident, delivered as a 22-page PDF to a distribution list. Nobody read it. Name the failure mode and prescribe a specific remedy that does not involve changing the analysis.

7.7 Recognizing failure. An analyst's product states: "Reporting confirms the facility is producing at capacity." The underlying basis is one trade press article whose own source is the company's marketing material. Name every problem in that sentence and rewrite it.

7.8 Recognizing failure. After an incident, a review finds that three separate teams each held one indicator, and no team held two. Name the failure mode and describe the organizational change that addresses it.

7.9 Improving collection. In the composite supply chain case from Section 1.6.5, write the standing requirement that would have prevented the failure, decomposed into EEIs and indicators, and state the review cadence.

7.10 Improving analysis. Take the 2002 Iraq case. You are given the same reporting and one week. Describe the three specific process changes you would make and what each would likely have changed in the final language.

7.11 Synthetic content. A video appears showing a public official making an inflammatory statement. It is single-source, single-angle, arrived within an hour, and aligns perfectly with one side of a contested dispute. Describe your verification sequence in order, and write the caveat you would attach if the sequence is inconclusive.

7.12 The ethical line. A colleague proposes accepting a document from a candidate who worked at a competitor, on the grounds that the candidate offered it voluntarily and it is "just context." Assess this. What do you do, in what order?

7.13 The ethical line. You are researching a fictional company for a legitimate business purpose and discover that its CEO's teenage child has a public social media account discussing family travel. Assess whether this is collectable, and state the rule you are applying.

7.14 Aggregation. You have located, from separate public sources, a private individual's employer, gym schedule, vehicle, neighborhood, and children's school. Each fact is individually public. Explain what you have created, whether the aggregate is public, and what obligation you now have regarding the material.

7.15 Confidence calibration. You have four independent indicators supporting a judgment, all from open sources, and no contradicting evidence. Your instinct is high confidence. Argue for moderate instead, then decide, and state the reason for the level you chose in one sentence suitable for a product.

Part 8: Hands-On Lab Exercises

General Lab Guidance

The three rules from the front of this chapter apply to every lab. No private individuals. Publicly available means genuinely public. Document as you go. Violating Rule 1 results in a zero for the lab regardless of analytic quality.

Rubric scale. Each criterion scores 0 to 4:

Score	Meaning
4	Professional quality; could be delivered to a real consumer unchanged
3	Competent; meets requirements with minor weaknesses
2	Developing; partial compliance, notable gaps in reasoning or documentation
1	Attempted; concept recognized but misapplied
0	Missing or fundamentally incorrect

Universal requirements, graded in every lab. Every judgment carries a confidence level. Every product names at least one intelligence gap. Every finding appears in a collection log meeting the Section 1.11.5 standard. Missing any of the three costs points in every lab.

Lab 1: Identifying Open Source Information in Everyday Life

Time required: 75 to 100 minutes **Format:** Individual **Difficulty:** Foundational

Objectives

1. Recognize the breadth of open source information available in ordinary environments.
2. Classify sources by category and assess their analytic value.

3. Apply the four tests for "open" to real sources.
4. Identify what a given source reveals that its publisher probably did not intend.
5. Assess your own organizational or personal exposure through the same lens.

Prerequisites

- Sections 1.1 and 1.11.5.
- Internet access for Part 2 only.
- No specialized tools.

Part 1: The Physical Environment Inventory (25 min)

Without using a computer, inventory open source information observable in one ordinary environment: a workplace lobby, a campus building, a shopping center, a public transit station, or a residential street. Do not photograph people. Do not enter non-public areas. Record at least fifteen items on Worksheet 1A.

Things students routinely miss and should look for: posted vendor and contractor signage, building directories, parking permits and reserved spaces, delivery schedules on loading dock signage, emergency evacuation diagrams showing internal layout, equipment nameplates and service stickers, wifi network names, badge reader models, camera makes and coverage angles, public meeting notices, permits posted in windows, recycling and waste contractor markings, and job postings on physical boards.

For each item, record what it reveals beyond its face content. A fire evacuation diagram in a lobby reveals internal layout. A service sticker on an HVAC unit reveals a vendor with facility access. A vendor sign reveals a supply relationship and a potential phishing pretext.

Part 2: The Digital Everyday Inventory (30 min)

Pick one **organization** as your subject: your school, your employer, a local government body, or a well-known company. Not a person. Using only public sources and no account creation, inventory at least fifteen items on Worksheet 1B across as many of these categories as you can:

Category	Look for
Own website	Leadership pages, org structure, locations, vendor logos, document downloads
Job postings	Technology stack, team structure, reporting lines, expansion signals
Public filings or records	Registration, licensing, permits, inspection reports, budget documents
Meeting records	Minutes, agendas, board packets, recorded public meetings
Procurement	Solicitations, awarded contracts, vendor lists

Category	Look for
Imagery	Publicly available satellite and street-level views of facilities
Press and trade coverage	Expansion, partnerships, incidents, disputes
Technical footprint	Public subdomains, certificate records, published documentation
Third-party mentions	Vendor case studies, conference talks, academic collaborations

For each item: source, URL, access date and time, and one sentence on what it reveals.

Part 3: Applying the Four Tests (15 min)

Select five items from your inventories, including at least one you felt uncertain about. Run each through the four tests on Worksheet 1C. For any item that fails a test, state which test and what you did about it.

Part 4: The Exposure Reversal (15 min)

Write a half-page assessment answering: **if an adversary compiled everything I just found, what could they do with it?** Name at least three specific capabilities the aggregate enables that no single item enables. Then name two reduction steps the organization could take, and one exposure that cannot reasonably be reduced.

Worksheet 1A: Physical Environment Inventory

#	Item observed	Location type	Category	What it reveals beyond its face content	Analytic value (H/M/L)
1					
...					
15					

Worksheet 1B: Digital Inventory

#	Item	Category	Source and URL	Access date/time	What it reveals	Value (H/M/L)
1						
...						
15						

Worksheet 1C: Four Tests

Item	Test 1 Access	Test 2 Identity	Test 3 Protection	Test 4 Purpose/harm	Proceed?	Notes

Expected Outcomes

- Thirty or more inventoried items across both parts, with documentation.
- At least three physical items whose analytic value exceeds their face content, most commonly evacuation diagrams, vendor signage, and equipment nameplates.
- Recognition that job postings are among the highest-value open sources about any organization, because they must be specific to work.
- At least one item where a test raised a genuine question. Students who report that all five items passed all four tests cleanly have usually chosen easy items; the rubric rewards engaging with a hard one.
- An exposure assessment that identifies aggregate capabilities, most commonly a credible phishing pretext, a physical access approach, and a schedule-based targeting opportunity.
- No collection about any identifiable individual. Names appearing incidentally on public leadership pages are acceptable; anything further is not.

Discussion Questions

1. Which single item in your inventory would you remove first if you were responsible for this organization's security, and what would it cost the organization to remove it?
2. Job postings are useful precisely because they are specific. How should an organization resolve that tension?
3. You inventoried a physical environment without a computer. What does the result suggest about the claim that OSINT equals internet research?
4. One of your exposures cannot reasonably be reduced. How should an organization think about irreducible exposure?

Grading Rubric: Lab 1 (40 points)

Criterion	Weight	4 points looks like
Physical inventory breadth and specificity	x2	15+ items across varied categories, specific rather than generic
"Reveals beyond face content"	x2	Consistently identifies second-

Criterion	Weight	4 points looks like
reasoning		order value, not restatement
Digital inventory breadth	x2	15+ items across five or more categories
Documentation quality	x2	Source, URL, date and time present for every digital item
Four tests application	x2	Applied honestly, including at least one genuinely difficult item
Exposure reversal	x1	Three aggregate capabilities named, specific and plausible
Reduction recommendations	x1	Actionable, with cost acknowledged
Rule compliance	x1	No collection about private individuals anywhere in the submission

Total possible: 52 weighted. Passing: 34.

Lab 2: Raw Information Versus Actionable Intelligence

Time required: 90 to 120 minutes **Format:** Individual **Difficulty:** Foundational to intermediate

Objectives

1. Distinguish open source data, open source information, and open source intelligence in practice.
2. Rate sources for reliability and claims for credibility as separate judgments.
3. Detect circular reporting in a provided source set.
4. Produce two products from one source set, one of which is OSIF and one of which is OSINT, and articulate the difference.

Prerequisites

- Sections 1.1.2, 1.11, and Chapter 1's source evaluation framework.
- No internet access required; all material is provided.

Scenario

You support a county emergency management office. A requirement has been issued:

PIR: Is the Fairmount Road bridge at risk of closure during the coming season, and if so, on what timeline?

Decision supported: Whether to pre-position an alternate route plan and notify the two school districts whose bus routes cross the bridge. Decision needed in 14 days.

Provided Reporting

#	Item	Provenance
1	State bridge inspection report, dated 14 months ago, rates the deck condition "poor" and the substructure "fair."	State transportation agency, published
2	County budget document lists Fairmount Road bridge rehabilitation as "unfunded, priority tier 2."	County published budget
3	A local newspaper article dated three weeks ago quotes an unnamed county employee saying the bridge "won't last another winter."	Local paper, one reporter, unnamed source
4	A regional news site published a story two weeks ago repeating the "won't last another winter" quote, citing the local paper.	Regional outlet, citing item 3
5	A television station's website published a story 12 days ago stating that county officials "have warned the bridge may fail," citing "reports."	TV station, source unclear
6	A community social media group has 60 posts about the bridge, most linking to items 3 through 5.	Public social media
7	Federal river gauge data shows the river 40% above seasonal median for the past six weeks.	Federal agency data feed
8	National weather outlook indicates above-normal precipitation probability for the coming season.	Federal agency published outlook
9	County commission meeting minutes from two months ago record a motion to seek grant funding for the bridge; the motion passed.	Published minutes

#	Item	Provenance
10	A grant program's published award list, issued last week, does not include the county.	Federal grant program, published
11	Street-level imagery of the bridge, captured 20 months ago, shows visible spalling on the deck edge and a temporary barrier on the north approach.	Commercial imagery, dated
12	A contractor's public project list includes "Fairmount Rd bridge deck assessment, completed" with a date four months ago.	Contractor website
13	A school district's published transportation plan shows two bus routes crossing the bridge, serving 340 students.	District published document
14	Social media post from an anonymous account claims "county already decided to close it in March."	Public social media, anonymous
15	Regional fuel prices rose 7% this quarter.	Trade publication

Step-by-Step Instructions

Step 1 (10 min). Restate the PIR in your own words, name the decision maker, and state the deadline. Write one sentence on what the decision maker will do differently depending on the answer.

Step 2 (15 min). Complete Worksheet 2A: classify each item as open source data, open source information, or knowledge. None is intelligence yet. Note which items are the *same claim* appearing more than once.

Step 3 (15 min). Complete Worksheet 2B: assign reliability (A to F) and credibility (1 to 6) to each item with a one-line justification. Pay particular attention to items 3, 4, 5, and 6.

Step 4 (15 min). Complete Worksheet 2C: the circular reporting audit. Trace the "won't last another winter" claim. How many items appear to corroborate it? How many are independent? What is the actual evidentiary weight of items 3 through 6 combined?

Step 5 (10 min). Identify the noise. At least two items do not bear on the PIR. Name them and explain why each is tempting.

Step 6 (15 min). Write **Product A: the OSIF summary**. One page. Organized, contextualized, sourced compilation of what is publicly known about the bridge. No judgment, no forecast, no

confidence level. This is deliberately the wrong product, and writing it is the point.

Step 7 (25 min). Write **Product B: the OSINT assessment**. Maximum one page, using the Chapter 1 template. Must include a BLUF judgment with a confidence level and estimative probability language, rated key facts, an alternative explanation considered and rejected, at least two assumptions with consequences, at least two ranked gaps, and at least three specific recommendations including at least one the office can execute this week.

Step 8 (10 min). Write a half-page comparison: what exactly is different between Product A and Product B, and what would happen if you delivered A to a decision maker who needed B?

Worksheet 2A: Ladder Classification

Item	OSD / OSIF / Knowledge	Duplicate of item #?	Why

Worksheet 2B: Source Evaluation

Item	Reliability	Credibility	Justification

Worksheet 2C: Circular Reporting Audit

Claim	Items carrying it	Genuinely independent items	Earliest identifiable source	Combined evidentiary weight

Expected Outcomes

- Items 15 and 12 should be triaged carefully: 15 is noise, and 12 is easy to dismiss but is actually relevant, because a completed deck assessment four months ago means a current engineering evaluation exists and can be requested. Students who identify item 12 as a collection opportunity rather than as noise are demonstrating the target skill.
- Items 3, 4, 5, and 6 should be recognized as one claim with one origin, not four corroborating sources. Their combined weight is that of a single unnamed source. Students who count them as four sources have made the error the lab is built to produce.
- Item 14 should be rated low on both scales and either excluded or explicitly caveated.
- Items 1, 2, 7, 8, 9, 10, 11, and 13 are the analytic backbone: an aged inspection with a poor deck rating, no funding, a failed grant, elevated river levels, an adverse seasonal outlook, and 340 students dependent on the crossing.
- Confidence should land at moderate. High is not defensible from a 14-month-old inspection and

an unnamed quote; low understates the convergence of independent official data.

- Recommendations should include requesting the four-month-old deck assessment and asking the state for a current inspection, because the strongest recommendation available is to close the most consequential gap rather than to act on the current picture.
- The Product A versus B comparison should conclude that A transfers the analytic burden to the reader.

Discussion Questions

1. Four items carried one claim. In a real environment, how would you detect this pattern quickly when you have 400 items instead of 15?
2. The strongest recommendation was to close a gap rather than to act. When is "collect more before deciding" the right answer, and when is it an analyst avoiding accountability?
3. Item 13 (340 students) does not help answer whether the bridge will fail. Why does it belong in the product anyway?
4. Your assessment carries moderate confidence. The consumer asks you to "just tell me yes or no." How do you respond?

Grading Rubric: Lab 2 (48 points)

Criterion	Weight	4 points looks like
Ladder classification and duplicate detection	x2	Accurate, and duplicates identified as duplicates
Source evaluation quality	x2	Ratings defensible and internally consistent
Circular reporting audit	x3	Correctly collapses items 3 through 6 to a single source's weight
Noise identification	x2	Correct, including recognizing item 12 as an opportunity rather than noise
Product A: correct OSIF discipline	x1	Genuinely withholds judgment; well organized and sourced
Product B: BLUF quality	x2	Judgment, confidence, estimative language in the opening
Product B: alternative explanation	x2	A real alternative developed and reasoned
Assumptions and gaps	x2	Two or more of each, with consequences and ranking
Recommendations	x2	Specific, assignable, time-bound, including a gap-closing action

Criterion	Weight	4 points looks like
Confidence calibration	x2	Moderate, with the reason stated
A versus B comparison	x2	Articulates the analytic burden transfer precisely

Total possible: 88 weighted. Passing: 60.

Lab 3: Historical Event Analysis Through an OSINT Lens

Time required: 2 to 3 hours **Format:** Individual, with optional group discussion **Difficulty:** Intermediate

Objectives

1. Reconstruct what was publicly knowable about a historical event before, during, and after it.
2. Classify the intelligence failure or success by failure mode.
3. Specify how better collection, analysis, or dissemination would have changed the outcome.
4. Distinguish hindsight bias from a defensible counterfactual.

Prerequisites

- Sections 1.6 and 1.7.
- Internet access.
- Completion of Labs 1 and 2.

Event Selection

Choose one event from List A, or propose an alternative meeting the rules below.

List A (recommended):

1. A major infrastructure failure with a documented prior warning record (a bridge collapse, a dam failure, a levee breach).
2. A corporate collapse or major fraud where public filings contained warning signs.
3. A large-scale product recall where public complaint data preceded the recall.
4. A natural disaster where forecast information existed before the response decisions were made.
5. A publicly documented data breach where exposed infrastructure had been externally visible.
6. A regulatory enforcement action where the underlying conduct was visible in public records for years.
7. A geopolitical surprise where open source indicators were later shown to have been present.

Rules for a proposed alternative: at least three years old, institutional rather than personal in nature, extensively documented in public sources including at least one formal review or inquiry, and not centered on identifying or characterizing a private individual.

Step-by-Step Instructions

Step 1 (20 min). Write the event summary: what happened, when, the consequences, and who the relevant decision makers were. Then write the counterfactual question in one sentence: *what decision, if made differently and earlier, would have changed the outcome most?*

Step 2 (30 min). Build the pre-event open source timeline on Worksheet 3A. Identify at least eight items of information that were publicly available **before** the event and bore on the counterfactual decision. For each: the source, its publication date, how far ahead of the event that was, and whether it was reasonably discoverable at the time. That last column matters, because judging 1998 by 2026 search capability is hindsight bias.

Step 3 (25 min). Complete Worksheet 3B: the failure mode analysis. Assess which of the five modes were present, with evidence for each. Assign a dominant mode and defend the assignment. Most events show several; the discipline is in ranking them.

Step 4 (25 min). Complete Worksheet 3C: the counterfactual intelligence product. Write the one-page assessment that **could** have been produced before the event, using only information from your Worksheet 3A timeline. Date it. Address it to the actual decision maker you named in Step 1. Include a confidence level that is honest given only the pre-event material, which usually means moderate at best.

This is the core of the lab. It is not a summary of what happened. It is the product that a competent analyst could have written at the time, and the confidence level has to reflect the pre-event evidence rather than your knowledge of the outcome.

Step 5 (20 min). Hindsight bias check on Worksheet 3D. For each item in your timeline, answer: would I have flagged this at the time, or does it only look significant because I know the ending? Then estimate how many *other* signals existed at the time that pointed nowhere. If you cannot name any, your analysis is probably suffering from hindsight bias, because real environments are full of indicators that lead nowhere.

Step 6 (25 min). Write the improvement analysis, one page: three specific changes to collection, analysis, or dissemination that would plausibly have changed the outcome. For each, name which failure mode it addresses, what it would have cost, and why it was not done at the time. That last part is the hardest and the most valuable, because the honest answer is usually structural rather than a matter of anyone being careless.

Step 7 (15 min). Write the transfer section, half a page: what does this event imply for an organization you know? Name one standing requirement you would write today as a result.

Worksheet 3A: Pre-Event Open Source Timeline

#	Information	Source and URL	Publication date	Days/ months before event	Discoverable at the time?	Bears on the counterfactual how?

Worksheet 3B: Failure Mode Analysis

Mode	Present?	Evidence	Rank
Collection			
Processing			
Analytic			
Dissemination			
Consumer			

Dominant mode and defense:

Worksheet 3C: Counterfactual Product

Use the Chapter 1 one-page template. Date it to a specific date before the event.

Worksheet 3D: Hindsight Bias Check

Timeline item	Would I have flagged it then?	Why or why not
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Signals present at the time that led nowhere:

Expected Outcomes

- Eight or more pre-event items with real sourcing and dates.
- An honest "discoverable at the time" column, including at least one item marked as not reasonably discoverable.
- A failure mode analysis that identifies multiple modes and ranks them, rather than defaulting to "they should have collected more." In most List A events, the dominant mode is analytic, dissemination, or consumer failure, not collection.
- A counterfactual product with confidence no higher than moderate, since pre-event evidence rarely supports more.
- A hindsight bias check that names at least two contemporaneous signals that led nowhere. Students who cannot find any have usually not looked at the period, only at the event.

- An improvement analysis where at least one proposed change is structural (a standing requirement, a fusion arrangement, a reporting line) rather than an exhortation to be more careful.

Discussion Questions

1. What was the dominant failure mode, and how confident are you in that ranking? What would change it?
2. Your counterfactual product carries moderate confidence. Would a decision maker have acted on it? If not, is that a failure of the product or of the consumer relationship?
3. How many contemporaneous signals led nowhere? What does that ratio imply about the realistic performance of any warning system?
4. Your improvement analysis named a change that was not made at the time. Why not? Give the structural answer, not the moral one.

Grading Rubric: Lab 3 (56 points)

Criterion	Weight	4 points looks like
Event selection compliance	x1	Meets all rules; institutional, well documented, formal review available
Counterfactual question precision	x2	One decision named, clearly the highest-leverage one
Pre-event timeline quality	x3	8+ items, fully sourced and dated, bearing clearly on the decision
"Discoverable at the time" honesty	x2	Genuinely assessed, including at least one negative
Failure mode analysis and ranking	x3	Multiple modes evidenced, dominant mode defended
Counterfactual product quality	x3	Written from pre-event material only, honest confidence, correctly addressed
Hindsight bias check	x3	Two or more dead-end signals identified; candid self-assessment
Improvement analysis	x2	Three changes, each mapped to a mode, with cost and reason it was not done
Structural insight	x2	At least one proposed change is structural rather than exhortative
Transfer to a known organization	x1	A specific standing requirement

Criterion	Weight	4 points looks like
		written
Documentation standard	x2	Every timeline item meets the Section 1.11.5 standard

Total possible: 96 weighted. Passing: 66.

Lab 4: Full OSINT Investigation of a Company

Time required: 2.5 to 3.5 hours **Format:** Individual **Difficulty:** Intermediate to advanced

Objectives

1. Conduct a structured, requirement-driven investigation of an organization using only public material.
2. Perform entity expansion before searching.
3. Separate facts from assessments throughout a full-length product.
4. Produce a professional OSINT summary with confidence levels, assumptions, gaps, and a documented collection log.

Prerequisites

- All of Part 1, particularly Sections 1.9 and 1.11.
- Completion of Labs 1 through 3.
- Chapter 1 product template.

Instructions: Two Tracks

Track A (primary): the provided fictional company. Work from the simulated public record packet in Section 8.4.1 below. This track exists because a fictional company cannot be researched live, and because it lets everyone be graded against the same material.

Track B (optional extension): a real organization. After completing Track A, repeat the method against a real **organization** of your choice: a public company, a university, a municipal government, or a nonprofit. Never an individual. Track B is where the skills become real, and it is graded as a bonus rather than as a requirement.

8.4.1 The Simulated Public Record Packet: Vireo Cold Chain Logistics

Everything below is fictional. Treat each item as though you found it in the stated source.

The requirement you have been given:

Client: a private investor considering a \$9M minority investment in Vireo Cold Chain

Logistics. **PIR:** Does the public record support Vireo's representation that it operates four temperature-controlled facilities, holds no material litigation, and has diversified customer concentration? **Deadline:** five business days.

Packet contents:

#	Source	Content
1	State business registry, State A	"Vireo Cold Chain Logistics, Inc." Active. Formed 6 years ago. Registered agent: a commercial agent service. Officers listed: two.
2	State business registry, State A	"Vireo CCL Holdings LLC." Active. Formed 6 years ago. Same registered agent address as item 1.
3	State business registry, State B	"Vireo Cold Storage II LLC." Active. Formed 3 years ago. One officer name matching an officer in item 1.
4	State business registry, State B	"Northridge Temp Control LLC." Inactive, administratively dissolved 18 months ago. Officer name matches an officer in item 1. Former name on file: "Vireo Cold Storage III LLC."
5	Company website, "Locations" page	Lists four facilities: State A metro (headquarters), State A secondary, State B north, State B south.
6	Company website, "About" page	Claims "over 40 enterprise clients." Names three publicly.
7	Commercial satellite imagery, State B south location	Current imagery shows a vacant lot with a concrete pad and no structure. Imagery from three years ago shows the same.
8	Commercial satellite imagery, State B north location	Shows an operating facility with 14 dock doors, 22 trailers in the yard, and rooftop refrigeration units.
9	County property records, State B south	The parcel matching the "State B south" address is owned by an unrelated party and zoned light industrial, undeveloped.

#	Source	Content
10	Building permit database, State B south county	A permit application for a cold storage structure was filed 26 months ago and marked "expired, no inspection."
11	Federal court docket search	One case: a breach of contract suit filed 9 months ago against "Vireo Cold Storage II LLC" by a former customer. Status: active. Amount claimed: \$2.1M.
12	State court docket, State B	Two small collections matters against "Northridge Temp Control LLC," both closed with judgments entered against the defendant.
13	UCC filing index, State A	Three filings: two by an equipment lessor covering refrigeration units, one by a lender covering "all assets," filed 14 months ago.
14	Job postings (aggregator)	Nine postings in the past 90 days, all for the State A metro and State B north locations. None for State A secondary or State B south.
15	Job postings (aggregator)	One posting for "Controller" at headquarters, listed twice in the past 7 months.
16	Professional networking profiles	The named CFO's profile lists "Vireo Cold Chain" with an end date two months ago. No successor listed on the company website.
17	Trade press article, 8 months ago	Quotes the CEO describing "our four-facility network" and an expansion plan.
18	Trade press article, 3 months ago	Reports that a large grocery distributor in the region consolidated cold chain vendors from five to two. Vireo is not named.
19	Customer public disclosure	One of the three publicly named clients issued a press release 4 months ago announcing a new

#	Source	Content
		logistics partner for its regional cold chain, naming a competitor.
20	Regulatory inspection database	Food safety inspection records exist for the State A metro and State B north facilities only. No records for the other two addresses.
21	Company website, archived version from 2 years ago	Locations page at that time listed three facilities. State B south appears for the first time 20 months ago.
22	Sanctions and watchlist screening (public lists)	No matches for the entities or named officers.
23	Local news, State B	An article 22 months ago reports a groundbreaking announcement for a new cold storage facility in the State B south county, quoting Vireo's CEO.
24	Utility service territory data (public)	The State A secondary address falls within a service territory; no further public detail available.

Steps:

Step 1 (15 min). Restate the PIR as three testable propositions, one per representation. Write, for each, what evidence would confirm it and what evidence would refute it. Do this before reading the packet closely a second time.

Step 2 (20 min). Entity expansion. Complete Worksheet 4A: every entity name, former name, jurisdiction, officer name, and shared address in the packet, with the relationships among them. Draw the structure as a text diagram. This step drives everything else.

Step 3 (30 min). Test proposition 1 (four facilities). Complete Worksheet 4B. For each claimed facility, list every packet item bearing on it, and reach a determination: operating, not operating, or unresolved. Note which sources are physical, which are documentary, and where they agree or conflict.

Step 4 (20 min). Test proposition 2 (no material litigation). Note what you can and cannot establish from docket searches alone, including which jurisdictions you would have searched and did not have access to.

Step 5 (20 min). Test proposition 3 (diversified customer concentration). Note carefully that the packet does not contain revenue data, and state what that means for your ability to answer. An honest "cannot be determined from public sources, here is what would resolve it" is a stronger answer than a guess.

Step 6 (20 min). Collection log. Complete Worksheet 4C for at least twelve packet items to the Section

1.11.5 standard, treating each stated source as the source of record.

Step 7 (40 min). Write the OSINT summary. Maximum two pages plus the collection log. Required sections: BLUF with confidence; findings organized by proposition, with facts and assessments visibly separated; entity structure summary; at least three assumptions with consequences; at least four ranked gaps; recommended next steps including which non-open sources the client should request in diligence; and a "what would change this assessment" section.

Step 8 (15 min). Write the ethics annex, half a page: which items in the packet involve information about identifiable individuals, how you handled them, and what you deliberately did not do.

Worksheet 4A: Entity Expansion

Entity name	Former names	Jurisdiction	Status	Officers	Shared indicators	Relationship to core entity
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Worksheet 4B: Facility Verification Matrix

Claimed facility	Documentary evidence	Physical evidence	Regulatory evidence	Hiring signal	Determination	Confidence
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Worksheet 4C: Collection Log

#	Claim	Source	Access date/time	Archived?	Independence	Rel.	Cred.	Verification performed	Initials
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Expected Outcomes

- **Facility finding.** State B south is assessed as not operating, with high confidence: three independent source types agree (imagery shows a vacant pad, property records show unrelated ownership, permit expired without inspection, and no inspection records or job postings exist). State A secondary is assessed as unresolved, with the specific gap named, since only utility territory data touches it and no inspection records or postings exist. State A metro and State B north are assessed as operating with high confidence.
- **The archived website finding.** Item 21 establishes that the four-facility claim postdates the groundbreaking announcement, which materially changes how item 17 reads. Students who use the archive item are demonstrating the target skill.
- **Litigation finding.** One active \$2.1M matter against a subsidiary, plus two closed judgments against a dissolved entity whose former name places it inside the Vireo family. The correct handling is to state that the representation is inconsistent with the record as to the active matter, without asserting intent, and to note that only two docket systems were searched.

- **Concentration finding.** Cannot be determined from the packet. Items 18 and 19 are adverse indicators, not evidence of concentration. The correct answer states the limit and specifies what to request: customer revenue schedules, aging reports, and top-customer contracts.
- **Key-person finding.** The CFO departure (item 16) plus the twice-listed Controller posting (item 15) plus the "all assets" lender filing (item 13) form a coherent finance-function concern that no single item supports. Students who connect these three are performing convergence reasoning correctly.
- **Entity expansion payoff.** The dissolved entity's former name is the thread that pulls the judgments into the family. A student who searches only "Vireo Cold Chain Logistics" misses items 4 and 12 entirely.
- **Honest gaps.** At least four, including customer concentration, State A secondary status, financial condition, and litigation in unsearched jurisdictions.
- **Ethics annex** that identifies the CFO and CEO as individuals appearing in a legitimate business context, handled by role and public statement only, with no personal research performed.

Discussion Questions

1. Entity expansion produced the single most consequential finding. Why is this step skipped so often in practice?
2. You could not determine customer concentration. Write two sentences to the client explaining that without sounding like you failed.
3. Three weak items (CFO departure, repeated Controller posting, all-assets lien) formed one finding. State the finding and the confidence level, and defend the level.
4. The archived website page changed how a CEO quotation reads. What does that suggest about archives as a source category?
5. Nothing in your product should allege intent. Where were you most tempted, and how did you word it instead?

Grading Rubric: Lab 4 (72 points)

Criterion	Weight	4 points looks like
Propositions restated as testable, with refutation criteria	x2	All three, with both confirming and refuting evidence specified
Entity expansion completeness	x3	All entities, former names, and shared indicators mapped; relationships drawn
Use of the former-name thread	x2	Item 4 correctly connected to item 12
Facility verification matrix	x3	All four assessed; source types distinguished; determinations

Criterion	Weight	4 points looks like
		defended
Use of the archived page	x2	Item 21 used to reframe item 17
Litigation finding precision	x2	Inconsistency stated without alleging intent; search limits named
Concentration finding honesty	x3	Correctly states it cannot be determined; specifies what would resolve it
Convergence reasoning on the finance signals	x2	Three items combined into one calibrated finding
Fact / assessment separation	x2	Visible throughout the product
Confidence calibration	x2	Different levels for different findings, each justified
Assumptions and ranked gaps	x2	Three-plus assumptions with consequences; four-plus ranked gaps
Recommended next steps	x2	Names the non-open sources to request in diligence
Collection log quality	x2	Twelve-plus items meeting the full standard
Ethics annex	x1	Individuals identified and handling explained
Length discipline	x1	Two pages plus log
Track B extension (bonus)	+4	Method correctly repeated against a real organization, no individuals researched

Total possible: 116 weighted, plus 4 bonus. Passing: 80.

Lab 5: Source Credibility and Corroboration on a Current Event

Time required: 2 to 3 hours **Format:** Individual **Difficulty:** Advanced

Objectives

1. Evaluate multiple live sources on the same event for reliability and credibility.
2. Trace claims toward their origin and detect circular reporting in real material.
3. Design and execute a corroboration plan.

4. Distinguish established, disputed, and unverified elements of a developing story.
5. Produce a calibrated assessment that resists the pressure to over-conclude.

Prerequisites

- Sections 1.11 and 2.4.
- Internet access.
- Completion of Labs 1 through 4.

Event Selection Rules

Choose an event that is: current (within the last 30 days), institutional or systemic (a policy action, an infrastructure or industrial incident, a market event, a public health development, a disaster, a regulatory action, a large-scale technical outage), and covered by at least six sources you can locate independently.

Do not choose: anything centered on a private individual; an active criminal matter where your analysis could read as an accusation; or an event where you cannot separate reporting from commentary. If your candidate event fails any of these, pick another.

A note on political topics. You may select a politically contested event, and doing so makes the lab more valuable. If you do, Step 6 is graded more strictly, and you must complete the opposing-case section in good faith.

Step-by-Step Instructions

Step 1 (15 min). Define the analytic question. Not "what happened," which is unanswerable this early, but a specific factual question with a yes/no or quantitative answer. Examples of the right shape: did a specific facility suspend operations; how many units were affected; did a named institution take a specific action; when did the outage begin. Name the decision maker who would need the answer and what they would do with it.

Step 2 (35 min). Build the source inventory on Worksheet 5A. At least eight sources, deliberately spanning: at least one primary source (an official statement, filing, dataset, or the affected organization's own communication), at least two mainstream outlets, at least one local or trade outlet, at least one source from a different country or language if available, and at least one social or user-generated source. For each: outlet, author if named, publication timestamp, what specific claims it makes about your question, and what its stated sourcing is.

Step 3 (30 min). Complete Worksheet 5B, the claim provenance matrix. Extract the four to six most important individual claims relevant to your question. For each claim, record which sources carry it, which appear to have independent knowledge, which cite another source, and the earliest identifiable appearance with a timestamp. Then classify each claim as **established** (multiple independent sources or a primary source), **disputed** (credible sources conflict), or **unverified** (single origin, uncorroborated).

This worksheet is the lab. Everything else supports it.

Step 4 (25 min). Corroboration plan and execution on Worksheet 5C. For each unverified or disputed claim, design a corroboration step using a *different type* of source than the ones already carrying the claim: a primary document, a dataset, imagery, a regulatory filing, a physical consistency check, or a source in a different jurisdiction. Execute at least three of these and record the result, including the ones that failed to resolve anything. Failed corroboration attempts are findings and are graded as such.

Step 5 (20 min). Source reliability assessment on Worksheet 5D. Rate each of your eight sources A through F, with justification based on track record, transparency about sourcing, corrections practice, and proximity to the event. Then rate each claim's credibility 1 through 6 separately. Note explicitly any case where a highly reliable source carries a low-credibility claim, and explain how you handled it.

Step 6 (25 min). Bias and opposing case on Worksheet 5E:

- Your initial belief about the answer, before Step 2.
- Three findings that ran against it, and honestly how you handled each.
- The strongest good-faith version of the interpretation you find least plausible, in at least five sentences. Not a caricature.
- What specific evidence would move you.

Step 7 (30 min). Write the assessment. Maximum two pages. Required: BLUF with confidence and estimative language; a table of claims classified established, disputed, and unverified; the corroboration results including failures; at least three assumptions with consequences; ranked gaps; what would change the assessment; and an explicit statement of how the story is likely to develop and what you would watch for.

Step 8 (10 min). Reliability appendix, one paragraph: where your sourcing is weakest and what a reader should discount as a result.

Worksheet 5A: Source Inventory

#	Outlet	Type	Author	Timestamp	Claims made	Stated sourcing
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Worksheet 5B: Claim Provenance Matrix

Claim	Sources carrying it	Apparently independent	Citing others	Earliest appearance and timestamp	Classification
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Worksheet 5C: Corroboration Plan and Results

Claim	Corroboration method (different source type)	Executed?	Result	Effect on classification
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Worksheet 5D: Reliability and Credibility

Source	Reliability (A-F)	Justification	Claim	Credibility (1-6)	Handling note
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Worksheet 5E: Bias Check

Prompt	Response
My initial belief	
Findings that contradicted it	
How I actually handled them	
Strongest good-faith opposing interpretation	
Evidence that would change my mind	

Expected Outcomes

- Eight or more sources spanning the required types, with timestamps.
- A provenance matrix in which a meaningful share of apparent corroboration collapses. This is the normal result for a 30-day-old event, and discovering it is the main learning outcome. Students who report that all sources were independent have usually not checked stated sourcing carefully.
- At least one claim classified as unverified that the student initially assumed was established.
- Three executed corroboration attempts, including at least one that failed to resolve the claim, reported as a finding rather than omitted.
- At least one case of a reliable source carrying a low-credibility claim, correctly handled by rating the two separately.
- Confidence lower than the student's Step 1 instinct. This is the expected direction of movement and is rewarded.
- An opposing case a holder of that view would accept as fair. This is the hardest criterion in the lab.
- A "watch for" section naming specific observable developments rather than general statements.

Discussion Questions

1. How much of what you believed at Step 1 survived the provenance matrix intact?
2. One corroboration attempt failed. Explain to a consumer why reporting the failure makes your assessment more useful rather than less.

3. Your confidence went down as you did more work. Give the two-sentence version of why that is a sign of competence.
4. Compare your opposing case to your final assessment. Did writing it change anything? If not, consider whether you wrote it before or after you had already concluded.
5. This event is 30 days old at most. What will you be able to say in six months that you cannot say now, and what does that imply about the value of dating and archiving your product?

Grading Rubric: Lab 5 (76 points)

Criterion	Weight	4 points looks like
Event selection compliance	x1	Meets all rules; institutional, adequately covered
Analytic question precision	x2	Specific, answerable, tied to a named decision maker
Source inventory breadth	x2	Eight-plus sources across all required types, with timestamps
Claim extraction quality	x2	Claims are discrete and testable, not summaries
Provenance tracing	x3	Independence versus citation correctly distinguished; earliest appearances established
Classification discipline	x3	Established, disputed, and unverified applied consistently and defensibly
Corroboration design	x3	Methods use genuinely different source types
Corroboration execution and honesty	x3	Three-plus executed; failures reported as findings
Reliability and credibility separation	x2	Rated independently; the mismatch case identified and handled
Bias check candor	x3	Contradicting findings engaged rather than dismissed
Opposing case quality	x3	Good-faith, substantive, recognizable to its holders
Assessment structure and BLUF	x2	Template followed; judgment and confidence first
Confidence calibration	x3	Defensible given the sourcing; overconfidence penalized heavily
Watch-for section	x1	Specific observables named

Criterion	Weight	4 points looks like
Reliability appendix	x1	Weakest sourcing named without hedging
Documentation standard	x2	Every source archived and logged per Section 1.11.5

Total possible: 144 weighted. Passing: 100.

Part 9: Capstone Assignment

The Sentinel Ridge Water District Investigation

Weight: 30% of chapter grade **Time required:** 5 to 7 hours **Format:** Individual, may be completed across multiple sittings **Deliverables:** Six items, listed in Section 9.4

9.1 The Scenario

You have been engaged as an OSINT analyst by the **Sentinel Ridge Water District (SRWD)**, a fictional public water utility serving 210,000 residents across two counties. Everything in this scenario is fictional.

About SRWD:

- Public agency governed by a five-member elected board. Subject to open meeting and public records law.
- 340 employees. Two treatment plants, 1,100 miles of distribution main, 14 pump stations.
- Annual operating budget approximately \$88M.
- No intelligence function. A three-person IT team. Security handled by facilities.

The situation. Three things are happening at once, which is normal and is part of the exercise.

Thread 1: the procurement decision. The board must vote in six weeks on a \$14M, eight-year contract to modernize the district's supervisory control and data acquisition (SCADA) systems. The apparent low bidder is **Arclight Municipal Systems**, a vendor SRWD has not worked with before. Arclight's proposal represents that it has completed eleven comparable municipal SCADA deployments, holds current certifications, has no material litigation, and maintains a 24-hour support operation staffed in-country.

Thread 2: the impersonation concern. Three district customers have forwarded emails appearing to come from SRWD's billing department, directing payment to a new portal. The district does not operate that portal. The general manager wants to know the scope.

Thread 3: the community concern. A local group has begun circulating a petition alleging that the SCADA modernization is a pretext for outsourcing operations, and that Arclight is connected to a firm involved in a service failure in another state. The board wants to know whether the factual claims in the petition are supported, so that it can respond accurately either way.

Your position. You report to the general manager. You have five business days. You have no budget for paid data services beyond ordinary public records fees.

9.2 The Raw Reporting

Twenty-eight items, unsorted and unevaluated. Some are decisive. Some are noise. At least two conflict with each other. At least one is probably false. At least one raises an ethical question you must handle explicitly rather than quietly.

#	Source	Content
1	Arclight proposal (provided by SRWD)	Claims eleven completed municipal SCADA deployments, current certifications, no material litigation, 24-hour in-country support.
2	State business registry, State X	"Arclight Municipal Systems LLC," active, formed 4 years ago. Registered agent: commercial service. Two managers listed.
3	State business registry, State X	"Arclight Industrial Controls LLC," administratively dissolved 2 years ago. One manager name matches item 2. Former name on file: "Kestrel Process Systems LLC."
4	Trade press archive, 5 years ago	An article describes a municipal water SCADA project failure in another state, resulting in a nine-day manual-operations period. The contractor named is "Kestrel Process Systems."
5	State utility commission proceeding record, 4 years ago	Public docket regarding the same failure. Findings reference inadequate change-control practice by the contractor. No sanction against individuals.
6	Arclight website, "Projects" page	Lists eleven municipal clients by name.

#	Source	Content
7	Public procurement portals, four states	Awarded-contract records confirm six of the eleven named clients. Two more appear as subcontracts under a different prime. Three cannot be located in any public award record.
8	Municipal meeting minutes, one of the three unconfirmed clients	Minutes from 3 years ago record a pilot agreement with Arclight, not a full deployment, and a subsequent decision not to proceed.
9	Federal court docket	One active case: a contract dispute filed 7 months ago by a municipal client against Arclight Municipal Systems LLC. Claim: \$1.4M.
10	State court docket, State X	Two closed collections matters against "Arclight Industrial Controls LLC." Judgments entered against the defendant.
11	Certification body public registry	Two of the three certifications claimed in the proposal are listed as current. The third shows as lapsed 11 months ago.
12	Job postings aggregator	Fourteen Arclight postings in the past 90 days. Twelve are for a location in a country other than the one represented as the support base. Two are domestic engineering roles.
13	Arclight website, careers page	States "global delivery model with 24/7 coverage."
14	Archived version of Arclight website, 14 months ago	The "Projects" page at that time listed seven municipal clients, not eleven.
15	Certificate transparency logs	Six certificates issued in the past 30 days for domains containing "sentinelridge" or close variants, none of which SRWD owns.
16	Domain registration data	Those six domains were registered across a nine-day window through two registrars, using privacy services.

#	Source	Content
17	Passive DNS	Four of the six domains have resolved to the same hosting address.
18	The forwarded customer emails (provided by SRWD)	Direct recipients to a payment portal on one of the six domains. Sender display name matches SRWD's billing address format.
19	Internet-wide scan data	SRWD's own address space shows an exposed remote management interface on a device at a pump station, reachable from the public internet.
20	Public code repository search	A configuration file containing an internal SRWD hostname and a hardcoded credential was committed to a public repository by an account whose profile lists a former SRWD contractor.
21	SRWD's own published board packets	Include a network diagram of the treatment plant control network, published as part of a capital request three years ago and still available on the district website.
22	Petition text (provided)	Alleges: (a) Arclight is "the same company" as the contractor in the out-of-state failure; (b) the contract "outsources operations"; (c) the board "concealed" the vendor's history.
23	Contract solicitation documents, SRWD (public)	Scope covers system modernization and support, and does not transfer operational control of the plants.
24	Board meeting minutes, SRWD	The vendor shortlist was discussed in open session; minutes record no discussion of vendor litigation history.
25	Social media	An account created three weeks ago has posted the petition claims 40 times across local groups, with near-identical phrasing. Two other accounts of

#	Source	Content
		similar age post the same text.
26	Local news	One article covering the petition, quoting a named organizer and a district spokesperson.
27	Professional networking profile	A profile for a named individual lists "Manager, Arclight Municipal Systems" and, previously, "Kestrel Process Systems," with dates spanning the failure period. The same profile lists personal information including family details and a home neighborhood.
28	Regional fuel and commodity report	Construction input costs rose 6% this quarter.

9.3 Guidance on the Substance

Not a hint about the answer, but a hint about the shape of a good answer.

A weak submission produces three separate lists of concerns, one per thread, and leaves the general manager to decide what matters. A strong submission recognizes that the threads have different urgency and different consumers, makes an explicit choice about what leads, and notices that one thread contains something requiring action today rather than in six weeks.

You will also find that the petition's three claims are not uniformly true or uniformly false. Assessing each separately, and saying so plainly, is the professional response. An analyst who validates all three or dismisses all three has not done the work.

Item 27 is in the packet on purpose. How you handle it is graded.

9.4 What You Must Produce

Deliverable 1: Classification and Evaluation Worksheet (15 points)

All 28 items in a table with:

- Classification: open source data, open source information, or knowledge.
- Reliability (A-F) and credibility (1-6), each with a one-line justification.
- Thread assignment: procurement, impersonation, community claims, own-exposure, or noise.
- Source category (registry, docket, imagery, technical, media, social, primary document).

Then, in one paragraph each: identify the two items that conflict and how you resolved the conflict; identify the item you assess as probably false and why; and identify the item that raises an ethical question and state how you handled it.

Deliverable 2: Source and Collection Plan (15 points)

Before analysis, produce the plan you would have written on day one:

- **Three PIRs**, one per thread, each tied to a specific decision and deadline.
- **For each PIR: three to five EEIs**, and for each EEI, two to four observable indicators.
- **A collection plan table**: each indicator assigned to a specific open source type, with the four tests from Section 1.1.4 assessed, cost, time, and whether it is collectable.
- **A prohibited-activity statement**: at least five specific things you will not do on this engagement, written in advance.
- **At least two gaps that open sources cannot close**, honestly identified, with a note on what non-open source would close each.

Deliverable 3: Lifecycle Map (15 points)

Map the **impersonation thread** through all six lifecycle phases with specific content, not descriptions of phases. Required in the analysis phase: name and apply one structured analytic technique. Required in the dissemination phase: state the consumer, format, channel, timing, and who must not receive it, plus the first paragraph as it would actually appear. Required in the feedback phase: four questions and a falsifiable accuracy test with a date.

Deliverable 4: Information Versus Intelligence Demonstration (10 points)

Take the procurement thread and produce two short products from the same material:

- **Product A (OSIF)**: a half-page organized, sourced compilation of what the public record shows about Arclight. No judgment, no confidence, no forecast.
- **Product B (OSINT)**: a half-page assessment with a judgment, confidence level, estimative language, and a recommendation.

Then, in one paragraph, state exactly what changed between them and what would happen if a board member received only Product A before the vote.

Deliverable 5: Professional OSINT Report (35 points)

The main deliverable. **Maximum three pages**, plus a collection log and an ethics annex which do not count against the limit. Addressed to the general manager, with the understanding that it will be read by the board.

Required structure:

1. **Bottom line**. Four sentences maximum. The single most important judgment, with a confidence

level and estimative language. Must make clear what requires action now versus what informs the six-week decision.

2. **Findings by thread.** For each of the three threads: facts (each with reliability and credibility ratings) visibly separated from assessments. For the community claims thread, assess each of the petition's three claims separately.
3. **Own-exposure findings.** Items 19, 20, and 21 concern SRWD itself. Treat them as their own section, because the consumer is the same person who can fix them.
4. **Entity structure summary.** The Arclight, Arclight Industrial Controls, and Kestrel relationship, stated carefully and without alleging intent.
5. **Key assumptions.** At least four, each with what changes if it is wrong.
6. **Intelligence gaps.** At least five, ranked by how much resolving each would change the assessment.
7. **Recommended actions.** At least seven, each specific, assignable to a named role, and time-bound. At least two must be executable this week at no cost. At least one must address a diligence question that only non-open sources can answer.
8. **What would change this assessment.** Two to four sentences.
9. **Collection log** meeting the Section 1.11.5 standard for at least eighteen items.
10. **Ethics annex**, one page maximum: every individual appearing anywhere in your work, how you handled each, what you deliberately did not collect, and your specific handling of item 27. Also state how you handled item 25, which concerns accounts rather than identified people.

Deliverable 6: Rationale Annex (10 points)

One page maximum:

- Why you selected your bottom-line judgment out of everything available.
- Why you chose your confidence level and not the level above or below it.
- Which three items you excluded that a less careful analyst would have included, and why.
- Which finding you were most tempted to overstate, and the wording you used instead.

9.5 Capstone Grading Rubric (100 points)

Deliverable	Criterion	Points
1. Classification	Classification accuracy across 28 items	4
	Reliability and credibility ratings defensible and consistent	4

Deliverable	Criterion	Points
	Thread assignment and noise identification correct	3
	Conflict identified and resolution reasoned	2
	Probable falsehood identified with reasoning	1
	Ethical item identified and handling stated	1
2. Source plan	Three PIRs, decision-linked and answerable	5
	EEIs sufficient and non-overlapping; indicators observable	4
	Four tests applied to each collection line	3
	Prohibited-activity statement written in advance	2
	Non-collectable gaps honestly identified	1
3. Lifecycle map	Planning and collection content specific	3
	Processing problems real, with consequences	3
	Analytic technique correctly applied, not merely named	4
	Dissemination design fits the consumer; opening paragraph strong	3
	Feedback questions and falsifiable accuracy test	2
4. Info vs intel	Product A genuinely withholds judgment	3
	Product B carries judgment, confidence, estimative language, recommendation	4
	Comparison paragraph articulates the difference precisely	3
5. Report	Bottom line: four sentences,	5

Deliverable	Criterion	Points
	judgment, confidence, now-versus-later distinction	
	Correct selection of the leading judgment	4
	Facts separated from assessments, all rated	4
	Petition claims assessed separately rather than as a bloc	4
	Own-exposure section present and actionable	3
	Entity structure stated carefully, no allegation of intent	3
	Assumptions with consequences	2
	Gaps ranked by analytic impact	3
	Recommendations specific, assignable, time-bound, including no-cost immediate actions	4
	Collection log: eighteen-plus items at full standard	2
	Ethics annex, including item 27 handling	4
	Length discipline	2
6. Rationale	Confidence defended against adjacent levels	4
	Exclusions identified and justified	3
	Temptation to overstate identified with the wording used instead	3
Total		100

Grade bands: 90 to 100 exemplary, 80 to 89 proficient, 70 to 79 competent, 60 to 69 developing, below 60 not yet passing.

Automatic deductions:

- Any judgment stated without a confidence level: minus 5 per instance.
- Any finding not traceable to a logged source: minus 3 per instance.

- No intelligence gaps named anywhere: minus 10.
 - Presenting an assessment as a fact: minus 5 per instance.
 - Alleging intent or wrongdoing beyond what the record supports: minus 8 per instance.
 - Collecting or reporting personal details about the individual in item 27 beyond their professional role and employment dates: minus 25. This is the ethical trap in the assignment, and the correct handling is to use the professional employment history, which is relevant to the entity relationship, and to exclude everything else, stating in the ethics annex that you did so and why.
-

Part 10: Answer Keys

Instructors may wish to remove this section from student copies.

10.1 Short Answer Key

1. OSINT is intelligence produced from publicly available, lawfully obtainable information collected without defeating protections. "Intelligence" does the most work, because it imposes the requirement, evaluation, analysis, judgment, and confidence obligations that distinguish a product from a search result. Many people can gather public material; the discipline is in what happens after gathering.
2. OSD is raw public material (an unedited livestream, a raw image). OSIF is collected and contextualized public material (a news article, a compiled filing list). OSINT is OSIF evaluated and analyzed against a requirement with a judgment and confidence level. The common error is delivering OSIF labeled as OSINT.
3. Access (could any member of the public obtain it lawfully), identity (can I obtain it without misrepresenting who I am), protection (am I defeating a control the owner placed), and purpose and harm (do I have a legitimate purpose, and is foreseeable harm proportionate). The fourth requires judgment rather than a legal determination, because aggregation of lawful items can produce an unlawful-feeling or harmful whole.
4. No, or at minimum not safely. Accidental exposure is not publication; the owner did not authorize access, and depending on jurisdiction, retrieving and using it may implicate computer misuse or trade secret law. The correct handling is to not collect it and, where appropriate, to notify the owner.
5. Acceptable answers draw from: wartime broadcast and press monitoring (institutionalized systematic content analysis); early Cold War (permanent government function, large translation programs, the status paradox); digitization and commercialization (commercial databases and imagery); the web (collapse of publication and retrieval cost); social and mobile (participants as publishers and sensors); the verification era (formalized geolocation and chronolocation, civilian investigative capability); the current period (mass open documentation plus generative content).
6. Credible internal assessments held that open sources supplied the majority of usable content in finished intelligence, while budget and institutional prestige concentrated on clandestine collection.

Reasonable answers note that formal standing has improved substantially since the mid-2000s, and that resourcing relative to contribution remains contested.

7. Publication cost collapse, consumer sensors, commercial imagery maturity, machine-readable public infrastructure, formalization of verification methods, and a public adversarially peer-reviewed community. Strong answers pick one and defend it; formalized verification is the most defensible choice because it made open source findings survivable under hostile scrutiny.

8. In the 1940s, obtaining a foreign broadcast was hard and, once obtained, its authenticity was rarely in question. Now the material arrives instantly and establishing that it depicts what it claims requires geolocation, chronolocation, provenance tracing, and corroboration. Collection became cheap; verification became the expensive step.

9. Any four of: cost, speed, shareability, accessibility, auditability, tipping and cueing, breadth. Shareability matters more than students assume because an unclassified finding can be used in court, in the press, with partners, and with the public, and a finding that cannot be acted on publicly is often worth less than a weaker one that can.

10. Any four of: volume, deception, circular reporting, availability bias, weak reach into intent, ephemerality. Absence of reporting is weak evidence of absence because open sources systematically under-represent disconnected populations, repressive environments, non-dominant languages, rural areas, and events that do not photograph well.

11. Open-source-first means exhausting publicly knowable information before tasking expensive collection. The value is in the narrowing step: it converts a broad, costly tasking into a precise, cheap one, so scarce collection is spent on the residual question rather than on material a browser could have supplied.

12. Two of: OSINT identifies where and when to image for GEOINT; supplies the background that makes HUMINT questions specific and testable; supplies identifiers that make signals interpretable; supplies corporate structure around a financial transaction; enriches attacker infrastructure for CTI; supplies ambient context for MASINT readings.

13. The liar's dividend is the advantage a bad actor gains once plausible fakes are common, because authentic evidence can be dismissed as fabricated. It makes verification harder than fake detection, since even a correct determination that content is authentic can be publicly disputed. The response is to shift from artifact hunting to provenance and independent corroboration.

14. Collection failure (remedy: new sources or disciplines), processing failure (translation, normalization, entity resolution, indexing), analytic failure (structured techniques, competing hypotheses, red teams), dissemination failure (format, channel, timing, sharing rules), consumer failure (relationship, framing, credibility, governance).

15. Any four of: assumption treated as fact (key assumptions check); layering (carry confidence forward and cite originals); source reliability failure (independent re-evaluation of sources regardless of fit); absence of alternatives (Analysis of Competing Hypotheses); confidence language exceeding evidence (published estimative scale, calibrated); dissent buried (surface disagreement in summary

judgments).

16. Missing controls: a requirement, a verification threshold, corroboration before naming anyone, explicit confidence language, and an accountable analyst. Publication made it irreversible because internet-scale distribution cannot be recalled and corrections never reach the same audience as the accusation, so the harm to a misidentified private person is permanent.

17. Anchoring in physical or documentary fact; genuinely independent sources; total sourcing transparency; specificity over generality; robustness under adversarial scrutiny; ethical grounding. Access was not a distinguishing feature because the techniques and material available in the success cases were also available in the failure cases; the difference was verification discipline and accountability.

18. Governments face constitutional and statutory limits on collecting and retaining information about their own populations, criminal predicate requirements, purpose limitation and retention schedules, protections for lawful political and religious activity, oversight bodies, transparency law, and increasingly policy limits on purchased data. Public availability of the source does not remove the collector's authority question.

19. The mosaic effect is the creation of a sensitive whole from individually non-sensitive public parts. "Each fact was public" is inadequate because the subject never published the aggregate, the aggregate enables harms no component enables, and the harm assessment attaches to what you created rather than to what you collected.

20. Claim, source name, URL, access date and time with time zone, archived copy, publisher type and apparent independence, reliability rating, credibility rating, verification steps performed, analyst initials. Without it, findings cannot be reproduced, circular reporting cannot be detected, ephemeral sources become unverifiable, and the product is an opinion rather than testable work.

10.2 Multiple Choice Key

#	Answer	Note
1	c	Lawful public access without circumvention
2	b	No judgment or confidence means OSIF
3	c	Misrepresentation of identity puts it outside the discipline
4	b	Broadcasts, publications, statistics, timetables
5	b	Recycled authentic content is the dominant deception form
6	b	Physical consistency is the strongest check when applicable
7	b	Public append-only issuance

#	Answer	Note
		record, often pre-operational
8	b	Transmission is expected; absence is the anomaly, not proof
9	c	Substantially available in nearly every documented case
10	c	Dissemination failure
11	b	Names, former names, subsidiaries, agents, jurisdictions
12	c	Verification of a counterparty before a transaction
13	b	Sensitive whole from non-sensitive public parts
14	b	Masked tiles indexed the concealed locations
15	b	Independence and cost of coordination, not quantity

10.3 True or False Key

#	Answer	If false, why
1	False	Accessibility is not authorization; privacy law, computer misuse law, terms of service, and harm considerations all still apply
2	False	OSINT means lawfully accessible, not free; imagery, records fees, and subscriptions are normal costs
3	True	
4	False	Reversed; governments generally face stricter constitutional, predicate, retention, and oversight constraints on collecting about individuals
5	False	In most documented failures the information was substantially available; the dominant failures were processing, analysis,

#	Answer	If false, why
		dissemination, or consumer failures
6	False	Metadata can be stripped or edited, so it is corroborating rather than conclusive
7	False	Open sources are systematically biased by connectivity, language, geography, and repression, so absence is weak evidence
8	False	Internet-scale publication is effectively irreversible; corrections do not reach the audience the accusation reached
9	True	
10	True	

10.4 Critical Thinking Exercise Guidance

7.1 Legally risky: informal social media screening of candidates may bring the activity within consumer reporting regulation, and may surface protected characteristics that create discrimination exposure. Ethically problematic: open-ended profiling of individuals, mosaic effect, and asymmetry of power. A defensible version: a documented, job-related verification limited to specified public items (professional credentials, licensure, public regulatory sanctions), performed uniformly across all candidates, through the organization's compliant process, with the candidate notified, and with a written scope that excludes personal social media.

7.2 OSINT can address capability and, partially, general targeting patterns, and it can fully address the district's own exposure. It cannot address the intent of a specific state actor toward a specific utility. Rewrite: "Which threat actor categories have publicly documented targeting of water utilities of our size in the past 24 months, which initial access methods did they use, and which of those methods are viable against our currently exposed assets? Needed before the security budget submission." Answerable, bounded, decision-linked.

7.3 The two established local accounts rate higher on reliability (perhaps B or C) than the two month-old single-topic accounts (F, unratable, with a note of possible coordination). Credibility of the claim itself is assessed separately and may be moderate given the local accounts. The timing pattern (20 minutes) is consistent with either a genuinely observed event or coordinated posting; the discriminating check is whether the two new accounts add independent detail or repeat phrasing. Repetition of phrasing collapses them to one source.

7.4 Suggested wording: "A document of unestablished provenance, posted anonymously, is internally consistent and consistent with other reporting; we assess its content as possibly accurate but cannot

corroborate its authenticity." To raise confidence: independent confirmation of a specific verifiable detail from the document against a separate record system, or attribution of the document to an identifiable custodian.

7.5 Three steps: (1) locate the named agency's original statement and read it directly, since one citation is traceable and the other nine are not; (2) determine whether "officials" in the nine refers to the same statement, by comparing exact figures and phrasing; (3) look for an independent count from a different source type (a facility, a registry, an international body) that did not derive from the official statement. Each step tests whether the eleven reports are one source or several.

7.6 Dissemination failure. Remedy without changing the analysis: a one-paragraph BLUF pushed through the channel the consumers actually read, with the 22-page document as an attachment; delivery to a named individual rather than a distribution list; and a confirmation-of-receipt step. Optionally, a standing agreement on what triggers an out-of-band notification.

7.7 Problems: "confirms" overstates a single source; the source is not independent because its sourcing is the subject's own marketing; "producing at capacity" is imprecise; no reliability or credibility rating; no confidence level; no gap named. Rewrite: "One trade press article, sourced to the company's own marketing material, describes the facility as operating at capacity. We assess this as possibly accurate but uncorroborated (single non-independent source, credibility 3). Independent corroboration would require shipping records, imagery of throughput indicators, or hiring data."

7.8 Processing and dissemination failure, with a structural cause: no fusion point. Remedy: a designated body or role responsible for integrating indicators across teams, a shared indicator repository, and a standing requirement that defines in advance what combinations constitute a warning. Note that exhorting teams to "communicate more" is not a remedy; the fix is structural.

7.9 Model answer. PIR: "Which single-source dependencies exist in our supply base, and what is each one's geographic hazard and financial exposure?" EEIs: identify sole-source components; locate each supplier's production facilities; assess natural hazard exposure at each location; assess each supplier's financial condition; identify qualified alternates. Indicators: supplier filings disclosing site concentration, public hazard maps, imagery of the facility, court and lien filings, credit signals, trade press. Review cadence: full refresh annually, exception review quarterly, and immediate review on any supplier change or regional event.

7.10 Three changes: (1) run Analysis of Competing Hypotheses with abandonment developed as a genuine alternative and evidence scored for disconfirmation, which would likely have surfaced that much reporting was consistent with both hypotheses; (2) run a key assumptions check on the central premise, which would have required the continuation premise to be defended rather than assumed; (3) re-evaluate source reliability independently of fit, and carry dissent into the summary judgments, which would likely have downgraded the stated confidence and made the disagreement visible to consumers.

7.11 Sequence: reverse image and video search for earlier appearances; identify the earliest publisher and timestamp; examine that account's history and network; seek a second angle or second publisher of the same moment; check for corroborating context (event schedule, other attendees' material, official records of the appearance); check physical consistency (setting, weather, shadows, audio

characteristics); check whether any authoritative source has addressed it. Caveat if inconclusive: "Content circulating widely; we cannot establish provenance, no independent second angle has been located, and the earliest identifiable appearance postdates the claimed event by X hours. We assess the content as unverified and make no judgment on its authenticity. Treat as unconfirmed."

7.12 Assess: the document is another company's protected material, and use exposes the organization to trade secret liability regardless of the analytic value. Order of actions: stop, do not read further or distribute; quarantine or return the material without copying; notify legal immediately; document the receipt and the handling; remove any findings derived from it and record the removal; ensure the candidate is advised of obligations through HR and legal, not by you. Note that voluntary provision by the candidate is not a defense.

7.13 Not collectable. The rule: no collection about individuals who are not themselves the legitimate subject of the requirement, and never about minors, regardless of public availability. The child is not a party to the business question. Even if the account were relevant, Test 4 (purpose and harm) fails decisively. Record in the ethics annex that the material was encountered and excluded.

7.14 You have created a profile enabling physical location, schedule prediction, and access to a minor's location. The aggregate is not public; the subject never published it and could not have consented to it. Obligations: do not extend it, do not share or publish it, delete it unless there is a documented lawful purpose and an authorized process (in which case escalate to that process rather than retaining it personally), and record what you did. This exercise is the mosaic effect made concrete.

7.15 The case for moderate: all four indicators are open source, so all four are exposed to the same systemic risks (deception, availability bias, circular derivation); absence of contradicting evidence in open sources is weak, since contradicting evidence is precisely what is least likely to be published; and no non-open corroboration exists. The case for high: four genuinely independent sources that an adversary could not cheaply coordinate is the standard convergence argument. The deciding question is independence: if the four trace to different source types and record systems, high is defensible; if any two share an origin, moderate. Model product sentence: "We assess with moderate confidence, based on four open source indicators from independent source types, with no non-open corroboration available and no contradicting reporting located; confidence would rise to high with corroboration from a source type outside open reporting."

Appendix A: Reusable Templates

A.1 Collection Log

#	Claim	Source name	URL	Access date/time (with TZ)	Archived copy	Publisher type	Independent?	Rel. (A-F)	Cred. (1-6)	Verification performed	Analyst
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A.2 The Four Tests Decision Record

PROPOSED COLLECTION: _____
REQUIREMENT IT SERVES: _____

TEST 1 ACCESS Obtainable lawfully by any member of the public? Y / N
TEST 2 IDENTITY Obtainable without misrepresenting who I am? Y / N
TEST 3 PROTECTION Am I defeating any control the owner placed? Y / N
TEST 4 PURPOSE/HARM Legitimate purpose, proportionate foreseeable harm? Y / N

ANY "NO" ON 1, 2, OR 4, OR ANY "YES" ON 3 → DO NOT COLLECT

DECISION: _____ BASIS: _____

A.3 Claim Provenance Matrix

Claim	Sources carrying it	Independent	Citing	Earliest appearance + timestamp	Classification (established / disputed / unverified)

A.4 Source Evaluation Quick Reference

Reliability: **A** completely reliable · **B** usually reliable · **C** fairly reliable · **D** not usually reliable · **E** unreliable · **F** cannot be judged

Credibility: **1** confirmed · **2** probably true · **3** possibly true · **4** doubtful · **5** improbable · **6** cannot be judged

Estimative language: almost certainly 95-99% · very likely 80-95% · likely 55-80% · roughly even 45-55% · unlikely 20-45% · very unlikely 5-20% · almost certainly not 1-5%

A.5 OSINT Product Skeleton

BOTTOM LINE (judgment + confidence + so-what)	3-4 sentences
FINDINGS: FACTS (each rated, each logged)	bullets
FINDINGS: ASSESSMENT (meaning, alternative considered)	1-2 paragraphs
KEY ASSUMPTIONS (each with consequence if wrong)	3-4 bullets
INTELLIGENCE GAPS (ranked by analytic impact)	3-5 bullets
RECOMMENDED ACTIONS (specific, assignable, dated)	3-7 bullets
WHAT WOULD CHANGE THIS ASSESSMENT	2-4 sentences
COLLECTION LOG (attached, does not count toward length)	
ETHICS ANNEX (individuals encountered and handling)	

A.6 Entity Expansion Checklist

Before searching any organization, enumerate:

- Legal name, trade names, and doing-business-as filings
- Former names, including pre-merger and pre-rebrand names
- Parent, subsidiaries, and affiliates

- Registered agent name and address, and other entities sharing it
 - Officer and manager names, and other entities listing them
 - Every jurisdiction of incorporation, operation, and property ownership
 - Domain names, brand names, and product names
 - Common misspellings and transliterations
-

Appendix B: Ethics and Self-Assessment Checklist

Confirm all twenty before submitting any product in this chapter.

Legality and authorization

1. Every source was lawfully accessible to any member of the public.
2. I did not misrepresent my identity anywhere.
3. I did not defeat any access control, paywall, or credential.
4. I did not accept material from anyone not authorized to share it.
5. I complied with terms of service where they applied.

Individuals 6. No private individual is the subject of my collection. 7. Any individual appearing incidentally appears only in a professional capacity relevant to the requirement. 8. No minors appear anywhere in my work. 9. No family members or associates of any subject were researched. 10. I have not created an aggregate profile of any person. 11. My ethics annex names every individual encountered and how I handled them.

Analytic quality 12. My first sentence contains a judgment and a confidence level. 13. Facts are visibly separated from assessments. 14. Every fact carries reliability and credibility ratings. 15. I traced key claims toward their origin and checked for circular reporting. 16. I considered at least one alternative explanation and said why I rejected it. 17. My assumptions are written down with their consequences. 18. My gaps are named and ranked. 19. I used estimative language from a published scale. 20. I did not allege intent or wrongdoing beyond what the record supports.

Appendix C: Researcher Operational Security

Brief, because it is easy to overdo and dangerous to ignore.

What is detectable. Website operators can see visits. Small organizations notice unusual patterns. Professional networking platforms may notify subjects that their profile was viewed. Some records systems log and disclose lookups. Automated collection is often detectable and frequently prohibited.

Reasonable practice.

- Use a research browser profile separate from your personal accounts, so you are not signed in to services that attribute activity to you.
- Do not create accounts that misrepresent your identity. If a source requires an account you cannot honestly hold, treat the source as unavailable.
- Assume the subject may learn of your interest, and be sure your collection would be defensible if they did.
- Archive at collection, both for verification and so you do not need to revisit a source repeatedly.
- Do not use employer infrastructure for research outside your authorized scope.
- Never use automated collection against a source whose terms prohibit it.

What OPSEC is not. It is not permission to be anonymous in order to do something you would not do openly. If concealment is what makes the collection possible, the collection is the problem, not the exposure.

Appendix D: Where to Go Next

Concepts to study after this chapter:

- Geolocation and chronolocation as practical skills, with structured practice on historical imagery.
- Corporate and beneficial ownership research across jurisdictions.
- Technical footprint analysis: certificate transparency, passive DNS, internet-wide scan data, and public code repositories.
- Maritime and aviation tracking as analytic domains, including their lawful limits.
- Structured analytic techniques in depth, particularly Analysis of Competing Hypotheses and key assumptions checks.
- Data protection law and computer misuse law as they apply in your jurisdiction, which is the most consequential unglamorous study you can do in this field.
- Verification of synthetic media, with an emphasis on provenance rather than artifact detection.

Practice to build now:

- Pick one public organization each week and produce a one-page OSINT summary with a confidence level. Date it. Score yourself in three months.
- Keep a forecast log. Nothing else improves calibration as cheaply.
- Perform your own exposure assessment quarterly, and treat the findings as a demonstration of what your methods can reach.
- Write the opposing case in good faith, every time. It remains the hardest skill in the discipline

and the clearest marker of an analyst rather than an advocate.

End of Chapter 2